

Relations and Functions

Chapter 1 · All Questions and Complete Solutions



Part 1 · Examples

Q

1.1

Empty and universal relations



Let A be the set of all students of a boys school.

Show that the relation R in A given by $R = \{(a, b): a \text{ is sister of } b\}$ is the empty relation and $R' = \{(a, b): \text{the difference between heights of } a \text{ and } b \text{ is less than 3 meters}\}$ is the universal relation.

Conclusion

Since the school is boys school, no student of the school can be sister of any student of the school.

Hence, $R = \varnothing$, showing that R is the empty relation.

It is also obvious that the difference between heights of any two students of the school has to be less than 3 meters.

This shows that $R' = A \times A$ is the universal relation.

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1.2

Equivalence relation - congruent triangles



Let T be the set of all triangles in a plane with R a relation in T given by

$R = \{(T_1, T_2): T_1 \text{ is congruent to } T_2\}$.

Show that R is an equivalence relation.

Conclusion

R is reflexive, since every triangle is congruent to itself.

Further, $(T_1, T_2) \in R \Rightarrow T_1 \text{ is congruent to } T_2 \Rightarrow T_2 \text{ is congruent to } T_1 \Rightarrow (T_2, T_1) \in R$.

Hence, R is symmetric.

Moreover, $(T_1, T_2), (T_2, T_3) \in R \Rightarrow T_1 \text{ is congruent to } T_2 \text{ and } T_2 \text{ is congruent to } T_3 \Rightarrow T_1 \text{ is congruent to } T_3 \Rightarrow (T_1, T_3) \in R$.

Therefore, R is an equivalence relation.

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1.3

Symmetric but not equivalence - perpendicular lines



Let L be the set of all lines in a plane and R be the relation in L defined as

$R = \{(L_1, L_2): L_1 \text{ is perpendicular to } L_2\}$.

Show that R is symmetric but neither reflexive nor transitive.

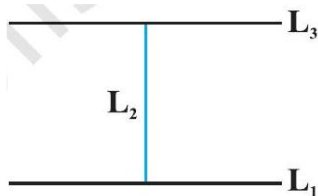


Fig 1.1

Given

R is not reflexive, as a line L_1 can not be perpendicular to itself, i.e., $(L_1, L_1) \notin R$.

R is symmetric as $(L_1, L_2) \in R$

Key formula

$\Rightarrow L_1$ is perpendicular to L_2

$\Rightarrow L_2$ is perpendicular to L_1

$\Rightarrow (L_2, L_1) \in R$.

Conclusion

R is not transitive.

Indeed, if L_1 is perpendicular to L_2 and

Fig 1.1

L_2 is perpendicular to L_3 , then L_1 can never be perpendicular to L_3 .

In fact, L_1 is parallel to L_3 , i.e., $(L_1, L_2) \in R, (L_2, L_3) \in R$ but $(L_1, L_3) \notin R$.

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1.4

Reflexive but
not
equivalence
relation



Show that the relation R in the set $\{1,2,3\}$ given by $R = \{(1,1), (2,2), (3,3), (1,2), (2,3)\}$ is reflexive but neither symmetric nor transitive.

Conclusion

R is reflexive, since $(1,1), (2,2)$ and $(3,3)$ lie in R .

Also, R is not symmetric, as $(1,2) \in R$ but $(2,1) \notin R$.

Similarly, R is not transitive, as $(1,2) \in R$ and $(2,3) \in R$ but $(1,3) \notin R$.

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1.5

Equivalence
relation -
divisibility
by 2



Show that the relation R in the set Z of integers given by $R = \{(a, b) : 2 \text{ divides } a-b\}$ is an equivalence relation.

Conclusion

R is reflexive, as 2 divides $(a-a)$ for all $a \in Z$.

Further, if $(a, b) \in R$, then 2 divides $a-b$.

Therefore, 2 divides $b-a$.

Hence, $(b, a) \in R$, which shows that R is symmetric.

Similarly, if $(a, b) \in R$ and $(b, c) \in R$, then $a-b$ and $b-c$ are divisible by 2. Now, $a-c = (a-b) + (b-c)$ is even (Why?

).

So, $(a-c)$ is divisible by 2.

This shows that R is transitive.

Thus, R is an equivalence relation in Z .

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1.6

Equivalence
relation -
odd/even
parity



Let R be the relation defined in the set $A = \{1,2,3,4,5,6,7\}$ by $R = \{(a, b) : \text{both } a \text{ and } b \text{ are either odd or even}\}$.

Show that R is an equivalence relation.

Further, show that all the elements of the subset $\{1,3,5,7\}$ are related to each other and all the elements of the subset $\{2, 4, 6\}$ are related to each other, but no element of the subset $\{1, 3, 5, 7\}$ is related to any element of the subset $\{2, 4, 6\}$.

Given

Take any element a in A .

Both a and a must be either odd or even, so $(a, a) \in R$.

Substitute

Next, if $(a, b) \in R \Rightarrow$ then both a and b must be either odd or even, so $\Rightarrow (b, a) \in R$.

Similarly, if $(a, b) \in R$ and $(b, c) \in R \Rightarrow$ then all elements a, b, c must be either even or odd at the same time, so $\Rightarrow (a, c) \in R$.

Hence, R is an equivalence relation.

Conclusion

All elements of $\{1,3,5,7\}$ are related to each other, since they are all odd.

In the same way, all elements of the subset $\{2, 4, 6\}$ are related to each other, since they are all even.

No element of $\{1, 3, 5, 7\}$ is related to any element of $\{2, 4, 6\}$, because elements of $\{1, 3, 5, 7\}$ are odd while elements of $\{2, 4, 6\}$ are even.

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1.7

One-one but
not onto -
roll numbers

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Let A be the set of all 50 students of Class X in a school.

Let $f: A \rightarrow N$ be function defined by $f(x) = \text{roll number of the student } x$.

Show that f is one-one but not onto.

Conclusion

No two different students of the class can have same roll number.

Therefore, f must be one-one.

We can assume without any loss of generality that roll numbers of students are from 1 to 50.

This implies that 51 in N is not roll number of any student of the class, so that 51 can not be image of any element of X under f .

Hence, f is not onto.

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1.8

One-one but
not onto -
 $f(x)=2x$ on N

✧

Show that the function $f: N \rightarrow N$, given by $f(x) = 2x$, is one-one but not onto.

Conclusion

The function f is one-one, for $f(x_1) = f(x_2) \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$.

Further, f is not onto, as for $1 \in N$, there does not exist any x in N such that $f(x) = 2x = 1$.

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1.9

Bijective
function -
 $f(x)=2x$ on R

✧

Prove that the function $f: R \rightarrow R$, given by $f(x) = 2x$, is one-one and onto.

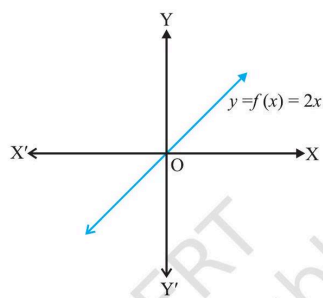


Fig 1.3

Conclusion

f is one-one, as $f(x_1) = f(x_2) \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$.

Also, given any real number y in R , there exists $\frac{y}{2}$ in R such that $f(\frac{y}{2}) = 2 \cdot (\frac{y}{2}) = y$.

Hence, f is onto.

Hence, f is onto.

Fig 1.3

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1.10

Onto but not
one-one
function

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Show that the function $f: N \rightarrow N$, given by $f(1) = f(2) = 1$ and $f(x) = x-1$, for every $x > 2$, is onto but not one-one.

Conclusion

f is not one-one, as $f(1) = f(2) = 1$.

But f is onto, as given any $y \in N$, $y \neq 1$, we can choose x as $y+1$ such that $f(y+1) = y+1-1 = y$.

Also for $1 \in N$, we have $f(1) = 1$.

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1.11

Neither one-one nor onto
- $f(x) = x^2$

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Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = x^2$, is neither one-one nor onto.

Conclusion

Since $f(-1) = 1 = f(1)$, f is not one-one.

Also, the element -2 in the co-domain $\mathbb{R}$ is not image of any element x in the domain $\mathbb{R}$ (Why?

).

Therefore f is not onto.

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1.12

Bijjective
piecewise
function

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Show that $f: \mathbb{N} \rightarrow \mathbb{N}$, given by

$x+1$, if x is odd,

$x-1$, if x is even

is both one-one and onto.

The image of 1 and -1 under f is 1.

Fig 1.4

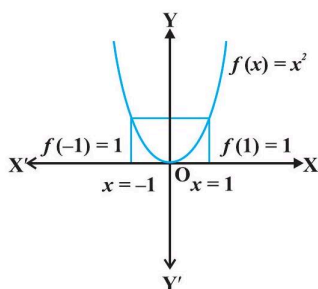


Fig 1.4

Conclusion

Suppose $f(x_1) = f(x_2)$.

If x_1 is odd and x_2 is even, then $x_1 + 1 = x_2 - 1$, i.e., $x_2 - x_1 = 2$, which is impossible.

In the same way, x_1 being even and x_2 being odd can also be ruled out, by a similar argument.

So both x_1 and x_2 must be either odd or even.

Suppose both x_1 and x_2 are odd.

Then $f(x_1) = f(x_2) \Rightarrow x_1 + 1 = x_2 + 1 \Rightarrow x_1 = x_2$.

Suppose instead both x_1 and x_2 are even.

Then also $f(x_1) = f(x_2) \Rightarrow x_1 - 1 = x_2 - 1 \Rightarrow x_1 = x_2$.

So f is one-one.

Also, any odd number $2r+1$ in the co-domain $\mathbb{N}$ is the image of $2r+2$ in the domain $\mathbb{N}$, and any even number $2r$ in the co-domain $\mathbb{N}$ is the image of $2r-1$ in the domain $\mathbb{N}$.

So f is onto.

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1.13

Onto implies
one-one on
finite set

✧

Show that an onto function $f: \{1,2,3\} \rightarrow \{1,2,3\}$ is always one-one.

Conclusion

Suppose f is not one-one.

Then there exists two elements, say 1 and 2 in the domain whose image in the co-domain is same.

Also, the image of 3 under f can be only one element.

Therefore, the range set can have at the most two elements of the co-domain $\{1,2,3\}$, showing that f is not onto, a contradiction.

Hence, f must be one-one.

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1.14

One-one
implies onto
on finite set

Show that a one-one function $f: \{1,2,3\} \rightarrow \{1,2,3\}$ must be onto.

Conclusion

Since f is one-one, three elements of $\{1,2,3\}$ must be taken to 3 different elements of the co-domain $\{1,2,3\}$ under f .

Hence, f has to be onto.

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1.15

Composition
of finite
functions

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Let $f: \{2,3,4,5\} \rightarrow \{3,4,5,9\}$ and $g: \{3,4,5,9\} \rightarrow \{7,11,15\}$ be functions defined as

$f(2) = 3, f(3) = 4, f(4) = f(5) = 5$ and $g(3) = g(4) = 7$ and $g(5) = g(9) = 11$.

Find $g \circ f$.

Conclusion

We have $g \circ f(2) = g(f(2)) = g(3) = 7, g \circ f(3) = g(f(3)) = g(4) = 7, g \circ f(4) = g(f(4)) = g(5) = 11$ and $g \circ f(5) = g(f(5)) = g(5) = 11$.

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1.16

gof not equal
to fog

Find $g \circ f$ and $f \circ g$, if $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ are given by $f(x) = \cos x$ and $g(x) = 3x^2$.

Show that $g \circ f \neq f \circ g$.

Conclusion

We have $g \circ f(x) = g(f(x)) = g(\cos x) = 3(\cos x)^2 = 3 \cos^2 x$.

Similarly, $f \circ g(x) = f(g(x)) = f(3x^2) = \cos(3x^2)$.

Note that $3 \cos^2 x \neq \cos(3x^2)$, for $x = 0$.

Hence, $g \circ f \neq f \circ g$.

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1.17

Invertible
function and
its inverse

Let $f: \mathbb{N} \rightarrow Y$ be a function defined as $f(x) = 4x+3$, where, $Y = \{y \in \mathbb{N}: y = 4x+3 \text{ for some } x \in \mathbb{N}\}$.

Show that f is invertible.

Find the inverse.

Given

Consider an arbitrary element y of Y .

By the definition of $Y, y = 4x+3$ for some x in the domain $\mathbb{N}$.

$$\text{So } x = \frac{(y-3)}{4}.$$

Define $g: Y \rightarrow \mathbb{N}$ by

$$g(y) = \frac{(y-3)}{4}. \text{ Now, } g \circ f(x) = g(f(x)) = g(4x+3) = \frac{(4x+3-3)}{4} = x \text{ and}$$

Conclusion

$$f \circ g(y) = f(g(y)) = f\left(\frac{(y-3)}{4}\right) = \frac{4(y-3)}{4} + 3 = y-3+3 = y.$$

This shows $g \circ f = I_{\mathbb{N}}$ and $f \circ g = I_Y$.

So f is invertible, and g is the inverse of f .

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1.18

Intersection
of
equivalence
relations

If R_1 and R_2 are equivalence relations in a set A , show that $R_1 \cap R_2$ is also an equivalence relation.

Conclusion

Since R_1 and R_2 are equivalence relations, $(a, a) \in R_1$, and $(a, a) \in R_2 \forall a \in A$.

This implies that $(a, a) \in R_1 \cap R_2, \forall a$, showing $R_1 \cap R_2$ is reflexive.

Further, $(a, b) \in R_1 \cap R_2 \Rightarrow (a, b) \in R_1$ and $(a, b) \in R_2 \Rightarrow (b, a) \in R_1$ and $(b, a) \in R_2 \Rightarrow$

$(b, a) \in R_1 \cap R_2$, hence, $R_1 \cap R_2$ is symmetric.

Similarly, $(a, b) \in R_1 \cap R_2$ and $(b, c) \in R_1 \cap R_2 \Rightarrow (a, c) \in R_1$ and $(a, c) \in R_2 \Rightarrow (a, c) \in R_1 \cap R_2$.

This shows that $R_1 \cap R_2$ is transitive.

Thus, $R_1 \cap R_2$ is an equivalence relation.

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1.19

Equivalence
relation on
ordered
pairs

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Let R be a relation on the set A of ordered pairs of positive integers defined by $(x, y) R(u, v)$ if and only if $xv = yu$.

Show that R is an equivalence relation.

Conclusion

Clearly, $(x, y) R(x, y)$, $\forall (x, y) \in A$, since $x \cdot y = y \cdot x$.

This shows that R is reflexive.

Further, $(x, y) R(u, v) \Rightarrow xv = yu \Rightarrow uy = vx$ and hence $(u, v) R(x, y)$.

This shows that R is symmetric.

Similarly, $(x, y) R(u, v)$ and $(u, v) R(a, b) \Rightarrow xv = yu$ and $ub = va \Rightarrow xv \cdot \frac{a}{u} = yu$

$$\frac{a}{u} \Rightarrow xv \cdot \frac{b}{v} = yu \cdot \frac{a}{u} \Rightarrow xb = ya \text{ and hence } (x, y) R(a, b).$$

Thus, R is transitive.

Thus, R is an equivalence relation.

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1.20

Equality of
two relations

✧

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Let R_1 be a relation in X given by $R_1 = \{(x, y): x-y \text{ is divisible by } 3\}$ and R_2 be another relation on X given by $R_2 = \{(x, y): \{x, y\} \subset \{1, 4, 7\} \text{ or } \{x, y\} \subset \{2, 5, 8\} \text{ or } \{x, y\} \subset \{3, 6, 9\}\}$.

Show that $R_1 = R_2$.

Conclusion

Note that the characteristic of sets $\{1, 4, 7\}$, $\{2, 5, 8\}$ and $\{3, 6, 9\}$ is that difference between any two elements of these sets is a multiple of 3.

Therefore, $(x, y) \in R_1 \Rightarrow x-y$ is a multiple of 3 $\Rightarrow \{x, y\} \subset \{1, 4, 7\}$ or $\{x, y\} \subset \{2, 5, 8\}$ or $\{x, y\} \subset \{3, 6, 9\} \Rightarrow (x, y) \in R_2$.

Hence, $R_1 \subset R_2$.

Similarly, $\{x, y\} \in R_2 \Rightarrow \{x, y\}$

$\subset \{1, 4, 7\}$ or $\{x, y\} \subset \{2, 5, 8\}$ or $\{x, y\} \subset \{3, 6, 9\} \Rightarrow x-y$ is divisible by 3 $\Rightarrow \{x, y\} \in R_1$.

This shows that $R_2 \subset R_1$.

Hence, $R_1 = R_2$.

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1.21

Equivalence
relation from
a function

✧

Let $f: X \rightarrow Y$ be a function.

Define a relation R in X given by $R = \{(a, b): f(a) = f(b)\}$.

Examine whether R is an equivalence relation or not.

Conclusion

For every $a \in X$, $(a, a) \in R$, since $f(a) = f(a)$, showing that R is reflexive.

Similarly, $(a, b) \in R \Rightarrow f(a) = f(b) \Rightarrow f(b) = f(a) \Rightarrow (b, a) \in R$.

Therefore, R is symmetric.

Further, $(a, b) \in R$ and $(b, c) \in R \Rightarrow f(a) = f(b)$ and $f(b) = f(c) \Rightarrow f(a) = f(c) \Rightarrow (a, c) \in R$, which implies that R is transitive.

Hence, R is an equivalence relation.

Q

1.22

Counting
one-one
functions

Find the number of all one-one functions from set $A = \{1,2,3\}$ to itself.

Conclusion

One-one function from $\{1, 2, 3\}$ to itself is simply a permutation on three symbols 1,2,3.

Therefore, total number of one-one maps from $\{1,2,3\}$ to itself is same as total number of permutations on three symbols 1,2,3 which is $3! = 6$.

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1.23

Counting
relations
with given
constraints

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Let $A = \{1,2,3\}$.

Then show that the number of relations containing (1,2) and (2,3) which are reflexive and transitive but not symmetric is three.

Given

The smallest relation R_1 containing (1,2) and (2,3) that is reflexive and transitive but not symmetric is $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$.

Substitute

If we add the pair (2,1) to R_1 , we get R_2 .

The relation R_2 is reflexive and transitive but not symmetric.

In the same way, adding (3,2) to R_1 gives the relation R_3 , with the same property.

We cannot add both pairs (2,1),(3,2), or just the single pair (3,1), to R_1 at the same time.

Doing so would force us to add the remaining pair to keep transitivity, and the relation would then also become symmetric, which we do not want.

Conclusion

So the total number of such relations is three.

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1.24

Counting
equivalence
relations

Show that the number of equivalence relation in the set $\{1,2,3\}$ containing (1,2) and (2,1) is two.

Given

The smallest equivalence relation R_1 containing (1,2) and (2,1) is $\{(1,1), (2,2),(3,3), (1,2),(2,1)\}$.

This leaves only 4 pairs: (2,3),(3,2), (1,3) and (3,1).

Substitute

Suppose we add just one of them, say (2,3), to R_1 .

Then symmetry forces us to also add (3,2), and transitivity then forces us to add (1,3) and (3,1) too.

Conclusion

So the only equivalence relation bigger than R_1 is the universal relation.

Hence, the total number of equivalence relations containing (1,2) and (2,1) is two.

Q

1.25

Sum of onto
functions not
onto

Consider the identity function $I_N: N \rightarrow N$ defined as $I_N(x) = x \forall x \in N$.

Show that although I_N is onto but $I_N + I_N: N \rightarrow N$ defined as

$(I_N + I_N)(x) = I_N(x) + I_N(x) = x + x = 2x$ is not onto.

Conclusion

Clearly I_N is onto.

But $I_N + I_N$ is not onto, as we can find an element 3 in the co-domain N such that there does not exist any x in the domain N with $(I_N + I_N)(x) = 2x = 3$.

Q**1.26**

Sum of one-
one functions
not one-one



Consider a function $f:[0, \frac{\pi}{2}] \rightarrow \mathbb{R}$ given by $f(x) = \sin x$ and $g:[0, \frac{\pi}{2}] \rightarrow \mathbb{R}$ given by $g(x) = \cos x$.

Show that f and g are one-one, but $f+g$ is not one-one.

Conclusion

Since for any two distinct elements x_1 and x_2 in $[0, \frac{\pi}{2}]$, $\sin x_1 \neq \sin x_2$ and $\cos x_1 \neq \cos x_2$, both f and g must be one-one.

But $(f+g)(0) = \sin 0 + \cos 0 = 1$ and $(f+g)(\frac{\pi}{2}) = \sin \frac{\pi}{2} + \cos \frac{\pi}{2} = 1$.

Therefore, $f+g$ is not one-one.

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Reflexive
symmetric
transitive -
five relations

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Determine whether each of the following relations are reflexive, symmetric and transitive:

(i) Relation R in the set $A = \{1, 2, 3, \dots, 13, 14\}$ defined as

$$R = \{(x, y) : 3x - y = 0\}$$

(ii) Relation R in the set N of natural numbers defined as

$$R = \{(x, y) : y = x + 5 \text{ and } x < 4\}$$

(iii) Relation R in the set $A = \{1, 2, 3, 4, 5, 6\}$ as

$$R = \{(x, y) : y \text{ is divisible by } x\}$$

(iv) Relation R in the set Z of all integers defined as

$$R = \{(x, y) : x - y \text{ is an integer}\}$$

(v) Relation R in the set A of human beings in a town at a particular time given by

(a) $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$

(b) $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$

(c) $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$

(d) $R = \{(x, y) : x \text{ is wife of } y\}$

(e) $R = \{(x, y) : x \text{ is father of } y\}$

Given

$$(i) R = \{(1, 3), (2, 6), (3, 9), (4, 12)\}$$

R is not reflexive because $(1, 1), (2, 2) \dots$ and $(14, 14) \notin R$.

R is not symmetric because $(1, 3) \in R$, but $(3, 1) \notin R$.

[since $3(3) \neq 0$].

R is not transitive because $(1, 3), (3, 9) \in R$, but $(1, 9) \notin R$. [$3(1) - 9 \neq 0$].

Hence, R is neither reflexive nor symmetric nor transitive.

Substitute

$$(ii) R = \{(1, 6), (2, 7), (3, 8)\}$$

R is not reflexive because $(1, 1) \notin R$.

R is not symmetric because $(1, 6) \in R$ but $(6, 1) \notin R$.

R is not transitive because there isn't any ordered pair in R such that $(x, y), (y, z) \in R$, so $(x, z) \notin R$.

Hence, R is neither reflexive nor symmetric nor transitive.

$$(iii) R = \{(x, y) : y \text{ is divisible by } x\}$$

We know that any number other than 0 is divisible by itself.

$$\text{Thus, } (x, x) \in R$$

So, R is reflexive.

$$(2, 4) \in R \text{ [because 4 is divisible by 2]}$$

$$\text{But } (4, 2) \notin R \text{ [since 2 is not divisible by 4]}$$

So, R is not symmetric.

$$\text{Let } (x, y) \text{ and } (y, z) \in R.$$

So, y is divisible by x and z is divisible by y.

$$\text{So, z is divisible by } x \Rightarrow (x, z) \in R$$

So, R is transitive.

So, R is reflexive and transitive but not symmetric.

$$(iv) R = \{(x, y) : x - y \text{ is an integer}\}$$

For $x \in Z, (x, x) \in R$ because $x - x = 0$ is an integer.

So, R is reflexive.

For $x, y \in Z$, if $x, y \in R$, then $x - y$ is an integer $\Rightarrow (y - x)$ is an integer.

$$\text{So, } (y, x) \in R$$

So, R is symmetric.

Let (x, y) and $(y, z) \in R$, where $x, y, z \in \mathbb{Z}$.

$\Rightarrow (x-y)$ and $(y-z)$ are integers.

$\Rightarrow x-z = (x-y) + (y-z)$ is an integer.

So, R is transitive.

So, R is reflexive, symmetric and transitive.

Conclusion

(v)

a) $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$

R is reflexive because $(x, x) \in R$

R is symmetric because,

If $(x, y) \in R$, then x and y work at the same place and y and x also work at the same place.

$(y, x) \in R$.

R is transitive because,

Let $(x, y), (y, z) \in R$

x and y work at the same place and y and z work at the same place.

Then, x and z also work at the same place.

$(x, z) \in R$.

Hence, R is reflexive, symmetric and transitive.

b) $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$

R is reflexive because $(x, x) \in R$

R is symmetric because,

If $(x, y) \in R$, then x and y live in the same locality and y and x also live in the same locality $(y, x) \in R$.

R is transitive because,

Let $(x, y), (y, z) \in R$

x and y live in the same locality and y and z live in the same locality.

Then x and z also live in the same locality.

$(x, z) \in R$.

Hence, R is reflexive, symmetric and transitive.

c) $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$

R is not reflexive because $(x, x) \notin R$.

R is not symmetric because,

If $(x, y) \in R$, then x is exactly 7 cm taller than y and y is clearly not taller than x .

$(y, x) \notin R$.

R is not transitive because,

Let $(x, y), (y, z) \in R$

x is exactly 7 cm taller than y and y is exactly 7 cm taller than z .

Then x is exactly 14 cm taller than z . $(x, z) \notin R$

Hence, R is neither reflexive nor symmetric nor transitive.

d) $R = \{(x, y) : x \text{ is wife of } y\}$

R is not reflexive because $(x, x) \notin R$.

R is not symmetric because,

Let $(x, y) \in R$, x is the wife of y and y is not the wife of x . $(y, x) \notin R$.

R is not transitive because,

Let $(x, y), (y, z) \in R$

x is wife of y and y is wife of z , which is not possible.

$$(x, z) \notin R.$$

Hence, R is neither reflexive nor symmetric nor transitive.

$$e) R = \{(x, y): x \text{ is father of } y\}$$

R is not reflexive because $(x, x) \notin R$.

R is not symmetric because,

Let $(x, y) \in R$, x is the father of y and y is not the father of x. $(y, x) \notin R$.

R is not transitive because,

$$\text{Let } (x, y), (y, z) \in R$$

x is father of y and y is father of z, x is not father of z. $(x, z) \notin R$.

Hence, R is neither reflexive nor symmetric nor transitive.

Q

2

Relation
neither
reflexive
symmetric
nor
transitive



Show that the relation R in the set R of real numbers, defined as $R = \{(a, b): a \leq b^2\}$ is neither reflexive nor symmetric nor transitive.

Conclusion

$$R = \{(a, b): a \leq b^2\}$$

$$\left(\frac{1}{2}, \frac{1}{2}\right) \notin R \text{ because } \frac{1}{2} > \left(\frac{1}{2}\right)^2$$

R is not reflexive.

$$(1, 4) \in R \text{ as } 1 < 4. \text{ But } 4 \text{ is not less than } 1^2.$$

$$(4, 1) \notin R$$

R is not symmetric.

$$(3, 2)(2, 1.5) \in R \text{ [Because } 3 < 2^2 = 4 \text{ and } 2 < (1.5)^2 = 2.25]$$

$$3 > (1.5)^2 = 2.25$$

$$\therefore (3, 1.5) \notin R$$

R is not transitive.

R is neither reflexive nor symmetric nor transitive.

Q

3

Relation $b = a+1$
properties



Check whether the relation R defined in the set $\{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b): b = a+1\}$ is reflexive, symmetric or transitive.

Conclusion

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$R = \{(a, b): b = a+1\}$$

$$R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$$

$$(a, a) \notin R, a \in A$$

$$(1, 1), (2, 2), (3, 3), (4, 4), (5, 5) \notin R$$

R is not reflexive.

$$(1, 2) \in R, \text{ but } (2, 1) \notin R$$

R is not symmetric.

$$(1, 2), (2, 3) \in R$$

$$(1, 3) \notin R$$

R is not transitive.

R is neither reflexive nor symmetric nor transitive.

Q

4

Relation $a \leq b$
properties



Show that the relation R in R defined as $R = \{(a, b): a \leq b\}$, is reflexive and transitive but not symmetric.

Conclusion

$$R = \{(a, b): a \leq b\}$$

$$(a, a) \in R$$

R is reflexive.

$$(2, 4) \in R \text{ (as } 2 < 4)$$

$$(4, 2) \notin R \text{ (as } 4 > 2)$$

R is not symmetric.

$$(a, b), (b, c) \in R$$

$$a \leq b \text{ and } b \leq c$$

$$\Rightarrow a \leq c$$

$$\Rightarrow (a, c) \in R$$

R is transitive.

R is reflexive and transitive but not symmetric.

Q

5

Relation $a \leq b^3$
properties

(())

Check whether the relation R in R defined by $R = \{(a, b): a \leq b^3\}$ is reflexive, symmetric or transitive.

Conclusion

$$R = \{(a, b): a \leq b^3\}$$

$$\left(\frac{1}{2}, \frac{1}{2}\right) \notin R, \text{ since } \frac{1}{2} > \left(\frac{1}{2}\right)^3$$

R is not reflexive.

$$(1, 2) \in R \text{ (as } 1 < 2^3 = 8)$$

$$(2, 1) \notin R \text{ (as } 2^3 > 1 = 8)$$

R is not symmetric.

$$\left(3, \frac{3}{2}\right), \left(\frac{3}{2}, \frac{6}{5}\right) \in R, \text{ since } 3 < \left(\frac{3}{2}\right)^3 \text{ and } \frac{2}{3} < \left(\frac{6}{5}\right)^3$$

$$\left(3, \frac{6}{5}\right) \notin R \text{ as } 3 > \left(\frac{6}{5}\right)^3$$

R is not transitive.

R is neither reflexive nor symmetric nor transitive.

Q

6

Symmetric
relation on
{1,2,3}



Show that the relation R in the set {1,2,3} given by $R = \{(1,2), (2,1)\}$ is symmetric but neither reflexive nor transitive.

Conclusion

$$A = \{1, 2, 3\}$$

$$R = \{(1, 2), (2, 1)\}$$

$$(1, 1), (2, 2), (3, 3) \notin R$$

R is not reflexive.

$$(1, 2) \in R \text{ and } (2, 1) \in R$$

R is symmetric.

$$(1, 2) \in R \text{ and } (2, 1) \in R$$

$$(1, 1) \in R$$

R is not transitive.

R is symmetric, but not reflexive or transitive.

Q

7

Equivalence
relation -
same page
count



Show that the relation R in the set A of all the books in a library of a college, given by $R = \{(x, y): x \text{ and } y \text{ have same number of pages}\}$ is an equivalence relation.

Conclusion

$$R = \{(x, y): x \text{ and } y \text{ have same number of pages}\}$$

R is reflexive since $(x, x) \in R$ as x and x have same number of pages.

R is reflexive.

$$(x, y) \in R$$

x and y have same number of pages and y and x have same number of pages $(y, x) \in R$

R is symmetric.

$$(x, y) \in R, (y, z) \in R$$

x and y have same number of pages, y and z have same number of pages.

Then x and z have same number of pages.

$$(x, z) \in R$$

R is transitive.

R is an equivalence relation.

Q

8

Equivalence
relation - $|a-b|$ even



Show that the relation R in the set $A = \{1, 2, 3, 4, 5\}$ given by

$R = \{(a, b) : |a-b| \text{ is even} \}$, is an equivalence relation.

Show that all the elements of $\{1, 3, 5\}$ are related to each other and all the elements of $\{2, 4\}$ are related to each other.

But no element of $\{1, 3, 5\}$ is related to any element of $\{2, 4\}$.

Given

$$a \in A$$

$$|a-a| = 0 \text{ (which is even)}$$

R is reflexive.

$$(a, b) \in R$$

$$\Rightarrow |a-b| \text{ is even}$$

$$\Rightarrow |-(a-b)| = |b-a| \text{ is even}$$

$$(b, a) \in R$$

R is symmetric.

$$(a, b) \in R \text{ and } (b, c) \in R$$

$$\Rightarrow |a-b| \text{ is even and } |b-c| \text{ is even}$$

$$\Rightarrow (a-b) \text{ is even and } (b-c) \text{ is even}$$

$$\Rightarrow (a-c) = (a-b) + (b-c) \text{ is even}$$

$$\Rightarrow |a-b| \text{ is even}$$

$$\Rightarrow (a, c) \in R$$

R is transitive.

R is an equivalence relation.

Conclusion

All elements of $\{1, 3, 5\}$ are related to each other because they are all odd.

So, the modulus of the difference between any two elements is even.

Similarly, all elements $\{2, 4\}$ are related to each other because they are all even.

No element of $\{1, 3, 5\}$ is related to any elements of $\{2, 4\}$ as all elements of $\{1, 3, 5\}$ are odd and all elements of $\{2, 4\}$ are even.

So, the modulus of the difference between the two elements will not be even.

Q

9

Equivalence
relation -
multiple of 4



Show that each of the relation R in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$, given by

(i) $R = \{(a, b) : |a-b| \text{ is a multiple of } 4\}$

(ii) $R = \{(a, b) : a = b\}$

is an equivalence relation.

Find the set of all elements related to 1 in each case.

Given

$$A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Substitute

i.

$$R = \{(a, b) : |a-b| \text{ is a multiple of } 4\}$$

$$a \in A, (a, a) \in R \text{ [} |a-a| = 0 \text{ is a multiple of } 4 \text{]}$$

R is reflexive.

$$(a, b) \in R \Rightarrow |a-b| \text{ is a multiple of } 4$$

$$\Rightarrow |-(a-b)| = |b-a| \text{ is a multiple of } 4$$

$$(b, a) \in R$$

R is symmetric.

$(a, b) \in R$ and $(b, c) \in R$
 $\Rightarrow |a-b|$ is a multiple of 4 and $|b-c|$ is a multiple of 4
 $\Rightarrow (a-b)$ is a multiple of 4 and $(b-c)$ is a multiple of 4
 $\Rightarrow (a-c) = (a-b) + (b-c)$ is a multiple of 4
 $\Rightarrow |a-c|$ is a multiple of 4
 $\Rightarrow (a, c) \in R$

R is transitive.

R is an equivalence relation.

The set of elements related to 1 is $\{1, 5, 9\}$ as

$|1-1| = 0$ is a multiple of 4.

$|5-1| = 4$ is a multiple of 4.

$|9-1| = 8$ is a multiple of 4.

Conclusion

ii.

$R = \{(a, b) : a = b\}$

$a \in A, (a, a) \in R$ [since $a = a$]

R is reflexive.

$(a, b) \in R$

$\Rightarrow a = b$

$\Rightarrow b = a$

$\Rightarrow (b, a) \in R$

R is symmetric.

$(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow a = b$ and $b = c$

$\Rightarrow a = c$

$\Rightarrow (a, c) \in R$

R is transitive.

R is an equivalence relation.

The set of elements related to 1 is $\{1\}$.

Q

10

Examples of relations with given properties



Give an example of a relation.

Which is

- (i) Symmetric but neither reflexive nor transitive.
- (ii) Transitive but neither reflexive nor symmetric.
- (iii) Reflexive and symmetric but not transitive.
- (iv) Reflexive and transitive but not symmetric.
- (v) Symmetric and transitive but not reflexive.

Given

i.

$A = \{5, 6, 7\}$

$R = \{(5, 6), (6, 5)\}$

$(5, 5), (6, 6), (7, 7) \notin R$

R is not reflexive as $(5, 5), (6, 6), (7, 7) \notin R$ $(5, 6), (6, 5) \in R$ and $(6, 5) \in R$, R is symmetric.

$\Rightarrow (5, 6), (6, 5) \in R$, but $(5, 5) \notin R$

R is not transitive.

Relation R is symmetric but not reflexive or transitive.

Substitute

ii.

$R = \{(a, b) : a < b\}$

$a \in R, (a, a) \notin R$ [since a cannot be less than itself]

R is not reflexive.

$(1, 2) \in R$ (as $1 < 2$)

But 2 is not less than 1

$\therefore (2, 1) \notin R$

R is not symmetric.

$$(a, b), (b, c) \in R$$

$$\Rightarrow a < b \text{ and } b < c$$

$$\Rightarrow a < c$$

$$\Rightarrow (a, c) \in R$$

R is transitive.

Relation R is transitive but not reflexive and symmetric.

iii.

$$A = \{4, 6, 8\}$$

$$A = \{(4, 4), (6, 6), (8, 8), (4, 6), (6, 8), (8, 6)\}$$

R is reflexive since $a \in A, (a, a) \in R$

R is symmetric since $(a, b) \in R$

$$\Rightarrow (b, a) \in R \text{ for } a, b \in R$$

R is not transitive since $(4, 6), (6, 8) \in R$, but $(4, 8) \notin R$

R is reflexive and symmetric but not transitive.

iv.

$$R = \{(a, b) : a^3 > b^3\}$$

$$(a, a) \in R$$

R is reflexive.

$$(2, 1) \in R$$

$$\text{But } (1, 2) \notin R$$

$\therefore$ R is not symmetric.

$$(a, b), (b, c) \in R$$

$$\Rightarrow a^3 > b^3 \text{ and } b^3 > c^3$$

$$\Rightarrow a^3 > c^3$$

$$\Rightarrow (a, c) \in R$$

$\therefore$ R is transitive.

R is reflexive and transitive but not symmetric

v.

$A = \{1, 3, 5\}$ Define a Relation R

On A.

$$R: A \rightarrow A$$

$$R = \{(1, 3), (3, 1), (1, 1), (3, 3)\}$$

Relation R is not Reflexive as $(5, 5) \notin R$

Relation R is Symmetric as

$$(1, 3) \in R \Rightarrow (3, 1) \in R$$

Relation R is Transitive

$$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$$

$$(3, 1) \in R, (1, 1) \in R \Rightarrow (3, 1) \in R$$

Conclusion

Alternative Answer

$R = (a, b)$: a is brother of b {suppose a and b are male}

Ref $\rightarrow$ a is not brother of a

So, $(a, a) \notin R$

Relation R is not Reflexive

Symmetric $\rightarrow$ a is brother of b so

b is brother of a

$$a, b \in R \Rightarrow (b, a) \in R$$

Transitive $\rightarrow$ a is brother of b and

b is brother of c so

a is brother of c

$$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$$

Q

11

Equivalence
relation -
equidistant
points



Show that the relation R in the set A of points in a plane given by $R = \{(P, Q) : \text{distance of the point } P \text{ from the origin is same as the distance of the point } Q \text{ from the origin}\}$, is an equivalence relation.

Further, show that the set of all points related to a point $P \neq (0,0)$ is the circle passing through P with origin as centre.

Given

$R = \{(P, Q) : \text{Distance of the point } P \text{ from the origin is same as the distance of the point } Q \text{ from the origin}\}$

Clearly, $(P, P) \in R$

R is reflexive.

$(P, Q) \in R$

Clearly R is symmetric.

$(P, Q), (Q, S) \in R$

$\Rightarrow$ The distance of P and Q from the origin is the same and also, the distance of Q and S from the origin is the same.

$\Rightarrow$ The distance of P and S from the origin is the same.

$(P, S) \in R$

R is transitive.

R is an equivalence relation.

Conclusion

The set of points related to $P \neq (0,0)$ will be those points whose distance from origin is same as distance of P from the origin.

Set of points forms a circle with the centre as origin and this circle passes through P .

Q

12

Equivalence
relation -
similar
triangles



Show that the relation R defined in the set A of all triangles as $R = \{(T_1, T_2) : T_1 \text{ is similar to } T_2\}$, is equivalence relation.

Consider three right angle triangles T_1 with sides 3, 4, 5, T_2 with sides 5, 12, 13 and T_3 with sides 6, 8, 10. Which triangles among T_1 , T_2 and T_3 are related?

Given

$R = \{(T_1, T_2) : T_1 \text{ is similar to } T_2\}$

R is reflexive since every triangle is similar to itself.

If $(T_1, T_2) \in R$, then T_1 is similar to T_2 .

T_2 is similar to T_1 .

$\Rightarrow (T_2, T_1) \in R$

R is symmetric.

$(T_1, T_2), (T_2, T_3) \in R$

T_1 is similar to T_2 and T_2 is similar to T_3 .

$\therefore T_1$ is similar to T_3 .

$\Rightarrow (T_1, T_3) \in R$

$\therefore R$ is transitive.

Conclusion

$$\frac{3}{6} = \frac{4}{8} = \frac{5}{10} = \left(\frac{1}{2}\right)$$

$\therefore$ Corresponding sides of triangles T_1 and T_3 are in the same ratio.

Triangle T_1 is similar to triangle T_3 .

Hence, T_1 is related to T_3 .

Q

13

Equivalence
relation -
polygon sides

()

Show that the relation R defined in the set A of all polygons as $R = \{(P_1, P_2) : P_1 \text{ and } P_2 \text{ have same number of sides}\}$, is an equivalence relation.

What is the set of all elements in A related to the right angle triangle T with sides 3, 4 and 5?

Given

$$R = \{(P_1, P_2) : P_1 \text{ and } P_2 \text{ have same number of sides}\}$$

$$(P_1, P_2) \in R \text{ as same polygon has same number of sides.}$$

$\therefore R$ is reflexive.

$$(P_1, P_2) \in R$$

$\Rightarrow P_1$ and P_2 have same number of sides.

$\Rightarrow P_2$ and P_1 have same number of sides.

$$\Rightarrow (P_2, P_1) \in R$$

$\therefore R$ is symmetric.

$$(P_1, P_2), (P_2, P_3) \in R$$

$\Rightarrow P_1$ and P_2 have same number of sides.

P_2 and P_3 have same number of sides.

$\Rightarrow P_1$ and P_3 have same number of sides.

$$\Rightarrow (P_1, P_3) \in R$$

$\therefore R$ is transitive.

R is an equivalence relation.

Conclusion

The elements in A related to right-angled triangle (T) with sides 3,4,5 are those polygons which have three sides.

Set of all elements in A related to triangle T is the set of all triangles.

Q

14

Equivalence
relation -
parallel lines



Let L be the set of all lines in XY plane and R be the relation in L defined as $R = \{(L_1, L_2) : L_1 \text{ is parallel to } L_2\}$.

Show that R is an equivalence relation.

Find the set of all lines related to the line $y = 2x + 4$.

Given

$$R = \{(L_1, L_2) : L_1 \text{ is parallel to } L_2\}$$

R is reflexive as any line L_1 is parallel to itself i.e., $(L_1, L_2) \in R$

If $(L_1, L_2) \in R$, then

$\Rightarrow L_1$ is parallel to L_2 .

$\Rightarrow L_2$ is parallel to L_1 .

$$\Rightarrow (L_2, L_1) \in R$$

$\therefore R$ is symmetric.

$$(L_1, L_2), (L_2, L_3) \in R$$

$\Rightarrow L_1$ is parallel to L_2

$\Rightarrow L_2$ is parallel to L_3

$\therefore L_1$ is parallel to L_3 .

$$\Rightarrow (L_1, L_3) \in R$$

$\therefore R$ is transitive.

R is an equivalence relation.

Conclusion

Set of all lines related to the line $y = 2x + 4$ is the set of all lines that are parallel to the line $y = 2x + 4$.

Slope of the line $y = 2x + 4$ is $m = 2$.

Line parallel to the given line is in the form $y = 2x + c$, where $c \in \mathbb{R}$.

Set of all lines related to the given line is given by $y = 2x + c$, where $c \in \mathbb{R}$.

Q

15

MCQ -
relation
properties on
{1,2,3,4}



Let R be the relation in the set {1,2,3,4} given by $R = \{(1,2), (2,2), (1,1), (4,4), (1,3), (3,3), (3,2)\}$.

Choose the correct answer.

- (A) R is reflexive and symmetric but not transitive.
 (B) R is reflexive and transitive but not symmetric.
 (C) R is symmetric and transitive but not reflexive.
 (D) R is an equivalence relation.

Conclusion

$$R = \{(1,2), (2,2), (1,1), (4,4), (1,3), (3,3), (3,2)\}$$

$$(a, a) \in R \text{ for every } a \in \{1,2,3,4\}$$

$\therefore$ R is reflexive.

$(1,2) \in R$ but $(2,1) \notin R$

$\therefore$ R is not symmetric.

$(a, b), (b, c) \in R$ for all $a, b, c \in \{1,2,3,4\}$

$\therefore$ R is not transitive.

R is reflexive and transitive but not symmetric.

The correct answer is B.

B //

Q

16

MCQ -
relation
membership
 $a=b-2$



Let R be the relation in the set N given by $R = \{(a, b): a = b-2, b > 6\}$.

Choose the correct answer.

- (A) $(2,4) \in R$
 (B) $(3,8) \in R$
 (C) $(6,8) \in R$
 (D) $(8,7) \in R$

Conclusion

$$R = \{(a, b): a = b-2, b > 6\}$$

Now,

$$b > 6, (2,4) \notin R$$

$$3 \neq 8-2$$

$$\therefore (3,8) \notin R \text{ and as } 8 \neq 7-2$$

$$\therefore (8,7) \notin R$$

Consider $(6,8)$

$$8 > 6 \text{ and } 6 = 8-2$$

$$\therefore (6,8) \in R$$

The correct answer is C.

C //

Q

17

One-one onto
- reciprocal
function

(())

Show that the function $f: \mathbb{R}^* \rightarrow \mathbb{R}^*$ defined by $f(x) = \frac{1}{x}$ is one-one and onto, where $\mathbb{R}^*$ is the set of all non-zero real numbers.

Is the result true, if the domain $\mathbb{R}^*$ is replaced by $\mathbb{N}$ with co-domain being same as $\mathbb{R}^*$?

Given

$$f: \mathbb{R}^* \rightarrow \mathbb{R}^* \text{ is by } f(x) = \frac{1}{x}$$

For one-one:

$$x, y \in \mathbb{R}^* \text{ such that } f(x) = f(y)$$

$$\Rightarrow \frac{1}{x} = \frac{1}{y}$$

$$\Rightarrow x = y$$

$\therefore f$ is one-one.

For onto:

$$\text{For } y \in \mathbb{R}^*, \text{ there exists } x = \frac{1}{y} \in \mathbb{R}^* [\text{as } y \neq 0] \text{ such that}$$

$$f(x) = \frac{1}{\left(\frac{1}{y}\right)} = y$$

$\therefore f$ is onto.

Given function f is one-one and onto.

Conclusion

$$\text{Consider function } g: \mathbb{N} \rightarrow \mathbb{R}^* \text{ defined by } g(x) = \frac{1}{x}$$

$$\text{We have, } g(x_1) = g(x_2) \Rightarrow \frac{1}{x_1} = \frac{1}{x_2} \Rightarrow x_1 = x_2$$

$\therefore g$ is one-one.

$$g \text{ is not onto as for } 1.2 \in \mathbb{R}^*, \text{ there exist any } x \text{ in } \mathbb{N} \text{ such that } g(x) = \frac{1}{1.2}$$

Function g is one-one but not onto.

Q

18

Injectivity
and
surjectivity
of x^2, x^3

⬡

Check the injectivity and surjectivity of the following functions:

(i) $f: \mathbb{N} \rightarrow \mathbb{N}$ given by $f(x) = x^2$

(ii) $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^2$

(iii) $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$

(iv) $f: \mathbb{N} \rightarrow \mathbb{N}$ given by $f(x) = x^3$

(v) $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^3$

Given

i.

$$\text{For } f: \mathbb{N} \rightarrow \mathbb{N} \text{ given by } f(x) = x^2$$

$$x, y \in \mathbb{N}$$

$$f(x) = f(y) \Rightarrow x^2 = y^2 \Rightarrow x = y$$

$\therefore f$ is injective.

$$2 \in \mathbb{N}.$$

$$\text{But, there does not exist any } x \text{ in } \mathbb{N} \text{ such that } f(x) = x^2 = 2$$

$\therefore f$ is not surjective

Function f is injective but not surjective.

Substitute

ii.

$f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^2$

$$f(-1) = f(1) = 1 \text{ but } -1 \neq 1$$

$\therefore f$ is not injective.

$-2 \in \mathbb{Z}$ But, there does not exist any $x \in \mathbb{Z}$ such that $f(x) = -2 \Rightarrow x^2 = -2$

$\therefore f$ is not surjective.

Function f is neither injective nor surjective.

iii.

$f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$

$$f(-1) = f(1) = 1 \text{ but } -1 \neq 1$$

$\therefore f$ is not injective.

$-2 \in \mathbb{Z}$ But, there does not exist any $x \in \mathbb{Z}$ such that $f(x) = -2 \Rightarrow x^2 = -2$

$\therefore f$ is not surjective.

Function f is neither injective nor surjective.

iv.

$f: \mathbb{N} \rightarrow \mathbb{N}$ given by $f(x) = x^3$

$$x, y \in \mathbb{N}$$

$$f(x) = f(y) \Rightarrow x^3 = y^3 \Rightarrow x = y$$

$\therefore f$ is injective.

$$2 \in \mathbb{N}.$$

But, there does not exist any x in $\mathbb{N}$ such that $f(x) = x^3 = 2$

$\therefore f$ is not surjective

Function f is injective but not surjective.

Conclusion

v.

$f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^3$

$$x, y \in \mathbb{Z}$$

$$f(x) = f(y) \Rightarrow x^3 = y^3 \Rightarrow x = y$$

$\therefore f$ is injective.

$$2 \in \mathbb{Z}.$$

But, there does not exist any x in $\mathbb{Z}$ such that $f(x) = x^3 = 2$

$\therefore f$ is not surjective.

Function f is injective but not surjective.

Q

19

Greatest
integer
function



Prove that the Greatest Integer Function $f: \mathbb{R} \rightarrow \mathbb{R}$, given by $f(x) = [x]$, is neither one-one nor onto, where $[x]$ denotes the greatest integer less than or equal to x .

Given

$f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = [x]$

Substitute

$$f(1.2) = [1.2] = 1, f(1.9) = [1.9] = 1$$

$$\therefore f(1.2) = f(1.9), \text{ but } 1.2 \neq 1.9$$

$\therefore f$ is not one-one.

Conclusion

Consider $0.7 \in \mathbb{R}$

$f(x) = [x]$ is an integer.

There does not exist any element $x \in \mathbb{R}$ such that $f(x) = 0.7$

$\therefore f$ is not onto.

The greatest integer function is neither one-one nor onto.

Q

20

Modulus
function



Show that the Modulus Function $f: \mathbb{R} \rightarrow \mathbb{R}$, given by $f(x) = |x|$, is neither one-one nor onto, where $|x|$ is x , if x is positive or 0 and $|x|$ is $-x$, if x is negative.

Key formula

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ is } f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$



Substitute

$$f(-1) = |-1| = 1 \text{ and } f(1) = |1| = 1$$

$$\therefore f(-1) = f(1) \text{ but } -1 \neq 1$$

$\therefore f$ is not one-one.

Conclusion

Consider $-1 \in \mathbb{R}$

$f(x) = |x|$ is non-negative.

There exist any element x in domain $\mathbb{R}$ such that $f(x) = |x| = -1$

$\therefore f$ is not onto.

The modulus function is neither one-one nor onto.

Q

21

Signum
function

(())

Show that the Signum Function $f: \mathbb{R} \rightarrow \mathbb{R}$, given by

1, if $x > 0$

0, if $x = 0$

-1, if $x < 0$

is neither one-one nor onto.

Given

$f: \mathbb{R} \rightarrow \mathbb{R}$ is



Key formula

$$\begin{aligned} &1, \text{ if } x > 0 \\ &0, \text{ if } x = 0 \\ &-1, \text{ if } x < 0 \end{aligned}$$



Substitute

$$f(1) = f(2) = 1, \text{ but } 1 \neq 2$$

$\therefore f$ is not one-one.

Conclusion

$f(x)$ takes only 3 values (1, 0, -1) for the element -2 in co-domain

$\mathbb{R}$, there does not exist any x in domain $\mathbb{R}$ such that $f(x) = -2$.

$\therefore f$ is not onto.

The signum function is neither one-one nor onto.

Q

22

One-one
function on
finite sets



Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and let $f = \{(1, 4), (2, 5), (3, 6)\}$ be a function from A to B .

Show that f is one-one.

Conclusion

$A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$

$f: A \rightarrow B$ is defined as $f = \{(1, 4), (2, 5), (3, 6)\}$

$$\therefore f(1) = 4, f(2) = 5, f(3) = 6$$

It is seen that the images of distinct elements of A under f are distinct.

$\therefore f$ is one-one.

Q

23

One-one,
onto or
bijective
functions

In each of the following cases, state whether the function is one-one, onto or bijective.

Justify your answer.

(i) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3-4x$

(ii) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1+x^2$

Given

i.

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3-4x$

$x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$

$$\Rightarrow 3-4x_1 = 3-4x_2$$

$$\Rightarrow -4x_1 = -4x_2$$

$$\Rightarrow x_1 = x_2$$

$\therefore f$ is one-one.

For any real number (y) in $\mathbb{R}$, there exists $\frac{3-y}{4}$ in $\mathbb{R}$ such that $f\left(\frac{3-y}{4}\right) = 3-4\left(\frac{3-y}{4}\right) = y$

$\therefore f$ is onto.

Hence, f is bijective.

Conclusion

ii.

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1+x^2$

$x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$

$$\Rightarrow 1+x_1^2 = 1+x_2^2$$

$$\Rightarrow x_1^2 = x_2^2$$

$$\Rightarrow x_1 = \pm x_2$$

$\therefore f(x_1) = f(x_2)$ does not imply that $x_1 = x_2$

Consider $f(1) = f(-1) = 2$

$\therefore f$ is not one-one.

Consider an element -2 in co domain $\mathbb{R}$.

It is seen that $f(x) = 1+x^2$ is positive for all $x \in \mathbb{R}$.

$\therefore f$ is not onto.

Hence, f is neither one-one nor onto.

Q

24

Bijective
function $A \times B$
to $B \times A$ 

Let A and B be sets.

Show that $f: A \times B \rightarrow B \times A$ such that $f(a, b) = (b, a)$ is bijective function.

Conclusion

$f: A \times B \rightarrow B \times A$ is defined as $(a, b) \mapsto (b, a)$.

$(a_1, b_1), (a_2, b_2) \in A \times B$ such that $f(a_1, b_1) = f(a_2, b_2)$

$$\Rightarrow (b_1, a_1) = (b_2, a_2)$$

$$\Rightarrow b_1 = b_2 \text{ and } a_1 = a_2$$

$$\Rightarrow (a_1, b_1) = (a_2, b_2)$$

$\therefore f$ is one-one.

$(b, a) \in B \times A$ there exist $(a, b) \in A \times B$ such that $f(a, b) = (b, a)$

$\therefore f$ is onto.

f is bijective.

Q

25

Bijection
piecewise
function
(garbled
source)

()

Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$ for all $n \in \mathbb{N}$.

State whether the function f is bijective.

Justify your answer.

Key formula

$$\frac{n+1}{2}, \text{ if } n \text{ is odd}$$

$$\frac{n}{2}, \text{ if } n \text{ is even}$$

Substitute

$$f(1) = \frac{1+1}{2} = 1 \text{ and } f(2) = \frac{2}{2} = 1$$

$$f(1) = f(2), \text{ where } 1 \neq 2$$

$\therefore f$ is not one-one.

Consider a natural number n in co domain $\mathbb{N}$.

Case I: n is odd

$\therefore n = 2r+1$ for some $r \in \mathbb{N}$ there exists $4r+1 \in \mathbb{N}$ such that

$$f(4r+1) = \frac{4r+1+1}{2} = 2r+1$$

Case II: n is even

$\therefore n = 2r$ for some $r \in \mathbb{N}$ there exists $4r \in \mathbb{N}$ such that

$$f(4r) = \frac{4r}{2} = 2r$$

$\therefore f$ is onto.

Conclusion

f is not a bijective function.

Q

26

One-one onto
- rational
function

Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$.

Consider the function $f: A \rightarrow B$ defined by $f(x) = \left(\frac{x-2}{x-3} \right)$.

Is f one-one and onto?

Justify your answer.

Given

$$A = \mathbb{R} - \{3\}, B = \mathbb{R} - \{1\} \text{ and } f: A \rightarrow B \text{ defined by } f(x) = \left(\frac{x-2}{x-3} \right)$$

$x, y \in A$ such that $f(x) = f(y)$

$$\Rightarrow \frac{x-2}{x-3} = \frac{y-2}{y-3}$$

$$\Rightarrow (x-2)(y-3) = (y-2)(x-3)$$

$$\Rightarrow x y - 3 x - 2 y + 6 = x y - 3 y - 2 x + 6$$

$$\Rightarrow -3 x - 2 y = -3 y - 2 x$$

$$\Rightarrow 3 x - 2 x = 3 y - 2 y$$

$$\Rightarrow x = y$$

$\therefore f$ is one-one.

Conclusion

Let $y \in B = \mathbb{R} - \{1\}$, then $y \neq 1$

The function f is onto if there exists $x \in A$ such that $f(x) = y$.

Now,

$$f(x) = y$$

$$\Rightarrow \frac{x-2}{x-3} = y$$

$$\Rightarrow x-2 = x y - 3 y$$

$$\Rightarrow x(1-y) = -3 y + 2$$

$$\Rightarrow x = \frac{2-3 y}{1-y} \in A [y \neq 1]$$

Thus, for any $y \in B$, there exists $\frac{2-3 y}{1-y} \in A$ such that

$$f\left(\frac{2-3y}{1-y}\right) = \frac{\left(\frac{2-3y}{1-y}\right)-2}{\left(\frac{2-3y}{1-y}\right)-3} = \frac{2-3y-2+2y}{2-3y-3+3y} = \frac{-y}{-1} = y$$

$\therefore f$ is onto.

Hence, the function is one-one and onto.

Q

27

MCQ - $f(x) = x^4$



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$.

Choose the correct answer.

- (A) f is one-one onto
- (B) f is many-one onto
- (C) f is one-one but not onto
- (D) f is neither one-one nor onto.

Given

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^4$

$x, y \in \mathbb{R}$ such that $f(x) = f(y)$

$$\Rightarrow x^4 = y^4$$

$$\Rightarrow x = \pm y$$

$\therefore f(x) = f(y)$ does not imply that $x = y$.

For example $f(1) = f(-1) = 1$

$\therefore f$ is not one-one.

Conclusion

Consider an element 2 in co domain $\mathbb{R}$ there does not exist any x in domain $\mathbb{R}$ such that $f(x) = 2$.

$\therefore f$ is not onto.

Function f is neither one-one nor onto.

The correct answer is D.

D //

Q

28

MCQ - $f(x) = 3x$



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 3x$.

Choose the correct answer.

- (A) f is one-one onto
- (B) f is many-one onto
- (C) f is one-one but not onto
- (D) f is neither one-one nor onto.

Given

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = 3x$

$x, y \in \mathbb{R}$ such that $f(x) = f(y)$

$$\Rightarrow 3x = 3y$$

$$\Rightarrow x = y$$

$\therefore f$ is one-one.

Conclusion

For any real number y in co domain $\mathbb{R}$, there exist $\frac{y}{3}$ in $\mathbb{R}$ such that $f\left(\frac{y}{3}\right) = 3\left(\frac{y}{3}\right) = y$

$$\frac{y}{3} = y$$

$\therefore f$ is onto.

Hence, function f is one-one and onto.

The correct answer is A.

A //

Q

29

One-one onto
function
 $x/(1+|x|)$

(())

Show that the function $f: \mathbb{R} \rightarrow \{x \in \mathbb{R} : -1 < x < 1\}$ defined by $f(x) = \frac{x}{1+|x|}$, $x \in \mathbb{R}$ is one one and onto function.

Given

$$f: \mathbb{R} \rightarrow \{x \in \mathbb{R} : -1 < x < 1\} \text{ is defined by } f(x) = \frac{x}{1+|x|}, x \in \mathbb{R}.$$

Substitute

For one-one:

$$f(x) = f(y) \text{ where } x, y \in \mathbb{R}$$

$$\Rightarrow \frac{x}{1+|x|} = \frac{y}{1+|y|}$$

If x is positive and y is negative,

$$\frac{x}{1+|x|} = \frac{y}{1+|y|}$$

$$\Rightarrow 2xy = x-y$$

Since, x is positive and y is negative,

$$x > y \Rightarrow x-y > 0$$

2xy is negative.

$$2xy \neq x-y$$

Case of x being positive and y being negative, can be ruled out.

$\therefore$ x and y have to be either positive or negative.

If x and y are positive,

$$f(x) = f(y)$$

$$\Rightarrow \frac{x}{1+x} = \frac{y}{1+y}$$

$$\Rightarrow x-x y = y-x y$$

$$\Rightarrow x = y$$

$$\therefore f \text{ is one-one.}$$

Conclusion

For onto:

Let $y \in \mathbb{R}$ such that $-1 < y < 1$.

If x is negative, then there exists $x = \frac{y}{1+y} \in \mathbb{R}$ such that

$$f(x) = f\left(\frac{y}{1+y}\right) = \frac{\left(\frac{y}{1+y}\right)}{1+\left|\frac{y}{1+y}\right|} = \frac{\frac{y}{1+y}}{1+\left(\frac{-y}{1+y}\right)} = \frac{y}{1+y-y} = y$$

If x is positive, then there exists $x = \frac{y}{1-y} \in \mathbb{R}$ such that

$$f(x) = f\left(\frac{y}{1-y}\right) = \frac{\left(\frac{y}{1-y}\right)}{1+\left|\frac{y}{1-y}\right|} = \frac{\frac{y}{1-y}}{1+\left(\frac{y}{1-y}\right)} = \frac{y}{1-y+y} = y$$

$\therefore$ f is onto.

Hence, f is one-one and onto.

Q

30

Injective
function x^3 

Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3$ is injective.

Conclusion

$f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^3$

For one-one:

$f(x) = f(y)$ where $x, y \in \mathbb{R}$

$x^3 = y^3 \dots \dots \dots (1)$

We need to show that $x = y$

Suppose $x \neq y$, their cubes will also not be equal.

$\Rightarrow x^3 \neq y^3$

This will be contradiction to (1).

$\therefore x = y$.

Hence, f is injective.

Q

31

Equivalence
relation on
power set

Given a non empty set X , consider $P(X)$ which is the set of all subsets of X .

Define the relation R in $P(X)$ as follows:

For subsets A, B in $P(X)$, $A R B$ if and only if $A \subset B$.

Is R an equivalence relation on $P(X)$?

Justify your answer.

Given

Since every set is a subset of itself, $A R A$ for all $A \in P(X)$.

$\therefore R$ is reflexive.

Substitute

Let $A R B \Rightarrow A \subset B$

This cannot be implied to $B \subset A$.

If $A = \{1, 2\}$ and $B = \{1, 2, 3\}$, then it cannot be implied that B is related to A .

$\therefore R$ is not symmetric.

If $A R B$ and $B R C$, then $A \subset B$ and $B \subset C$.

$\Rightarrow A \subset C$

$\Rightarrow A R C$

$\therefore R$ is transitive.

Conclusion

R is not an equivalence relation as it is not symmetric.

Q

32

Counting
onto
functions

Find the number of all onto functions from the set $\{1, 2, 3, \dots, n\}$ to itself.

Conclusion

Onto functions from the set $\{1, 2, 3, \dots, n\}$ to itself is simply a permutation on n symbols $1, 2, 3, \dots, n$.

Thus, the total number of onto maps from $\{1, 2, 3, \dots, n\}$ to itself is the same as the total number of permutations on n symbols $1, 2, 3, \dots, n$, which is $n!$.

Q

33

Equal
functions
check

Let $A = \{-1, 0, 1, 2\}$, $B = \{-4, -2, 0, 2\}$ and $f, g: A \rightarrow B$ be functions defined by $f(x) = x^2 - x$,

$x \in A$ and $g(x) = 2 \left| x - \frac{1}{2} \right| - 1$, $x \in A$.

Are f and g equal?

Justify your answer.

(Hint: One may note that two functions $f: A \rightarrow B$ and $g: A \rightarrow B$ such that $f(a) = g(a)$

$\forall a \in A$, are called equal functions).

Conclusion

It is given that $A = \{-1, 0, 1, 2\}$, $B = \{-4, -2, 0, 2\}$

Also, $f, g: A \rightarrow B$ is defined by $x^2 - x, x \in A$ and $g(x) = 2 \mid x - \frac{1}{2} \mid - 1, x \in A$.

$$f(-1) = (-1)^2 - (-1) = 1 + 1 = 2$$

$$g(-1) = 2 \mid (-1) - \frac{1}{2} \mid - 1 = 2 \left(\frac{3}{2} \right) - 1 = 3 - 1 = 2$$

$$\Rightarrow f(-1) = g(-1)$$

$$f(0) = (0)^2 - 0 = 0$$

$$g(0) = 2 \mid 0 - \frac{1}{2} \mid - 1 = 2 \left(\frac{1}{2} \right) - 1 = 1 - 1 = 0$$

$$\Rightarrow f(0) = g(0)$$

$$f(1) = (1)^2 - 1 = 0$$

$$g(1) = 2 \mid 1 - \frac{1}{2} \mid - 1 = 2 \left(\frac{1}{2} \right) - 1 = 1 - 1 = 0$$

$$\Rightarrow f(1) = g(1)$$

$$f(2) = (2)^2 - 2 = 2$$

$$g(2) = 2 \mid 2 - \frac{1}{2} \mid - 1 = 2 \left(\frac{3}{2} \right) - 1 = 3 - 1 = 2$$

$$\Rightarrow f(2) = g(2)$$

$$\therefore f(a) = g(a) \forall a \in A$$

Hence, the functions f and g are equal.

Q

34

MCQ -
reflexive
symmetric
not transitive
count



Let $A = \{1, 2, 3\}$.

Then number of relations containing (1,2) and (1,3) which are reflexive and symmetric but not transitive is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Conclusion

The given set is $A = \{1, 2, 3\}$.

The smallest relation containing (1,2) and (1,3) which are reflexive and symmetric but not transitive is given by,

$$R = \{(1,1), (2,2), (3,3), (1,2), (1,3), (2,1), (3,1)\}$$

This is because relation R is reflexive as $\{(1,1), (2,2), (3,3)\} \in R$.

Relation R is symmetric as $\{(1,2), (2,1)\} \in R$ and $\{(1,3), (3,1)\} \in R$.

Relation R is transitive as $\{(3,1), (1,2)\} \in R$ but $(3,2) \notin R$.

Now, if we add any two pairs (3,2) and (2,3) (or both) to relation R , then relation R will become transitive.

Hence, the total number of desired relations is one.

The correct answer is A.

A //

Q

35

MCQ -
equivalence
relations
containing
(1,2)



Let $A = \{1, 2, 3\}$.

Then number of equivalence relations containing (1,2) is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Conclusion

The given set is $A = \{1, 2, 3\}$.

The smallest equivalence relation containing (1,2) is given by;

$$R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$$

Now, we are left with only four pairs i.e., (2,3), (3,2), (1,3) and (3,1).

If we add any one pair [say $(2,3)$] to R_1 , then for symmetry we must add $(3,2)$.

Also, for transitivity we are required to add $(1,3)$ and $(3,1)$.

Hence, the only equivalence relation (bigger than R_1) is the universal relation.

This shows that the total number of equivalence relations containing $(1,2)$ is two.

The correct answer is B.

B //

Q

EX1.3

-1

Composition
gof of finite
functions



Let $f: \{1, 3, 4\} \rightarrow \{1, 2, 5\}$ and $g: \{1, 2, 5\} \rightarrow \{1, 3\}$ be given by $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(1, 3), (2, 3), (5, 1)\}$.

Write down gof .

Conclusion

The functions $f: \{1, 3, 4\} \rightarrow \{1, 2, 5\}$ and $g: \{1, 2, 5\} \rightarrow \{1, 3\}$ are $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(1, 3), (2, 3), (5, 1)\}$

$$g \circ f(1) = g[f(1)] = g(2) = 3 \text{ [as } f(1) = 2 \text{ and } g(2) = 3]$$

$$g \circ f(3) = g[f(3)] = g(5) = 1 \text{ [as } f(3) = 5 \text{ and } g(5) = 1]$$

$$\text{gof}(4) = g[f(4)] = g(1) = 3 \text{ [as } f(4) = 1 \text{ and } g(1) = 3]$$

$$\therefore \text{gof} = \{(1, 3), (3, 1), (4, 3)\}$$

Q

EX1.3

-2

Composition
distributes
over sum and
product

()

Let f, g, h be functions from R to R .

Show that

$$(f+g) \circ h = f \circ h + g \circ h$$

$$(f \cdot g) \circ h = (f \circ h) \cdot (g \circ h)$$

Given

$$(f+g) \circ h = f \circ h + g \circ h$$

$$\text{LHS} = [(f+g) \circ h](x)$$

$$= (f+g)[h(x)] = f[h(x)] + g[h(x)]$$

$$= (f \circ h)(x) + g \circ h(x)$$

$$= \{(f \circ h) + (g \circ h)\}(x) = \text{RHS}$$

$$\therefore \{(f+g) \circ h\}(x) = \{(f \circ h) + (g \circ h)\}(x) \text{ for all } x \in R$$

Hence, $(f+g) \circ h = f \circ h + g \circ h$

Conclusion

$$(f \cdot g) \circ h = (f \circ h) \cdot (g \circ h)$$

$$\text{LHS} = [(f \cdot g) \circ h](x)$$

$$= (f \cdot g)[h(x)] = f[h(x)] \cdot g[h(x)]$$

$$= (f \circ h)(x) \cdot (g \circ h)(x)$$

$$= \{(f \circ h) \cdot (g \circ h)\}(x) = \text{RHS}$$

$$\therefore [(f \cdot g) \circ h](x) = \{(f \circ h) \cdot (g \circ h)\}(x) \text{ for all } x \in R$$

Hence, $(f \cdot g) \circ h = (f \circ h) \cdot (g \circ h)$

Q

EX1.3

-3

gof and fog -
modulus and
cube root



Find $g \circ f$ and $f \circ g$, if

i.

$$f(x) = |x| \text{ and } g(x) = |5x-2|$$

ii.

$$f(x) = 8x^3 \text{ and } g(x) = x \frac{1}{3}$$

Given

i.

$$f(x) = |x| \text{ and } g(x) = |5x-2|$$

$$\therefore g \circ f(x) = g(f(x)) = g(|x|) = |5|x|-2|$$

$$f \circ g(x) = f(g(x)) = f(|5x-2|) = |5x-2| = |5x-2|$$

Conclusion

ii.

$$f(x) = 8x^3 \text{ and } g(x) = x \frac{1}{3}$$

$$\therefore g \circ f(x) = g(f(x)) = g(8x^3) = (8x^3) \frac{1}{3} = 2x$$

$$f \circ g(x) = f(g(x)) = f\left(x \frac{1}{3}\right) = 8\left(x \frac{1}{3}\right)^3 = 8x$$

Q

EX1.3

-4

Self-inverse
rational
function

If $f(x) = \frac{(4x+3)}{(6x-4)}$, $x \neq \frac{2}{3}$, show that $f \circ f(x) = x$, for all $x \neq \frac{2}{3}$.

What is the reverse of f ?

Conclusion

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) = f\left(\frac{4x+3}{6x-4}\right) \\&= \frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4} = \frac{16x+12+18x-12}{24x+18-24x-16} = \frac{34x}{34} = x\end{aligned}$$

$$\therefore f \circ f(x) = x \text{ for all } x \neq \frac{2}{3}$$

$$\Rightarrow f \circ f = 1$$

Hence, the given function f is invertible and the inverse of f is f itself.

Q

EX1.3

-5

Checking
existence of
inverse -
three
functions

State with reason whether the following functions have inverse.

i.

$f: \{1,2,3,4\} \rightarrow \{10\}$ with $f = \{(1,10), (2,10), (3,10), (4,10)\}$

ii.

$g: \{5,6,7,8\} \rightarrow \{1,2,3,4\}$ with $g = \{(5,4), (6,3), (7,4), (8,2)\}$

iii.

$h: \{2,3,4,5\} \rightarrow \{7,9,11,13\}$ with $h = \{(2,7), (3,9), (4,11), (5,13)\}$

Given

i.

$f: \{1,2,3,4\} \rightarrow \{10\}$ with $f = \{(1,10), (2,10), (3,10), (4,10)\}$

f is a many one function as $f(1) = f(2) = f(3) = f(4) = 10$

$\therefore f$ is not one-one.

Function f does not have an inverse.

Substitute

ii.

$g: \{5,6,7,8\} \rightarrow \{1,2,3,4\}$ with $g = \{(5,4), (6,3), (7,4), (8,2)\}$

g is a many one function as $g(5) = g(7) = 4$

$\therefore g$ is not one-one.

Function g does not have an inverse.

Conclusion

iii.

$h: \{2,3,4,5\} \rightarrow \{7,9,11,13\}$ with $h = \{(2,7), (3,9), (4,11), (5,13)\}$

All distinct elements of the set $\{2,3,4,5\}$ have distinct images under h .

$\therefore h$ is one-one.

h is onto since for every element y of the set $\{7,9,11,13\}$, there exists an element x in the set $\{2,3,4,5\}$, such that $h(x) = y$.

h is a one-one and onto function.

Function h has an inverse.

Q

EX1.3

-6

Inverse of a
rational
function on
[-1,1]

Show that $f: [-1,1] \rightarrow \mathbb{R}$, given by $f(x) = \frac{x}{(x+2)}$ is one-one.

Find the inverse of the function $f: [-1,1] \rightarrow \text{Range } f$.

(Hint: For $y \in \text{Range } f$, $y = f(x) = \frac{x}{x+2}$, for some x in $[-1,1]$, i. e., $x = \frac{2y}{(1-y)}$)

Given

$f: [-1,1] \rightarrow \mathbb{R}$, given by $f(x) = \frac{x}{(x+2)}$

For one-one

$$\begin{aligned}
 f(x) &= f(y) \\
 \Rightarrow \frac{x}{x+2} &= \frac{y}{y+2} \\
 \Rightarrow x y+2 x &= x y+2 y \\
 \Rightarrow 2 x &= 2 y \\
 \Rightarrow x &= y
 \end{aligned}$$

$\therefore f$ is a one-one function.

It is clear that $f: [-1, 1] \rightarrow \mathbb{R}$ is onto.

$f: [-1, 1] \rightarrow \mathbb{R}$ is one-one and onto and therefore, the inverse of the function $f: [-1, 1] \rightarrow \mathbb{R}$ exists.

Substitute

Let $g: \text{Range } f \rightarrow [-1, 1]$ be the inverse of f .

Let y be an arbitrary element of range f .

Since $f: [-1, 1] \rightarrow \text{Range } f$ is onto, we have:

$y = f(x)$ for some $x \in [-1, 1]$

$$\begin{aligned}
 \Rightarrow y &= \frac{x}{x+2} \\
 \Rightarrow x y+2 y &= x \\
 \Rightarrow x(1-y) &= 2 y \\
 \Rightarrow x &= \frac{2 y}{1-y}, y \neq 1
 \end{aligned}$$

Now, let us define $g: \text{Range } f \rightarrow [-1, 1]$ as

$$g(y) = \frac{2 y}{1-y}, y \neq 1$$

Conclusion

Now,

$$\begin{aligned}
 (g \circ f)(x) &= g(f(x)) = g\left(\frac{x}{x+2}\right) = \frac{2\left(\frac{x}{x+2}\right)}{1-\frac{x}{x+2}} = \frac{2 x}{x+2-x} = \frac{2 x}{2} = x \\
 (f \circ g)(y) &= f(g(y)) = f\left(\frac{2 y}{1-y}\right) = \frac{\frac{2 y}{1-y}}{\frac{2 y}{1-y}+2} = \frac{2 y}{2 y+2-2 y} = \frac{2 y}{2} = y \\
 \therefore g \circ f &= I_{[-1, 1]} \text{ and } f \circ g = I_{\text{Range } f} \\
 \therefore f^{-1} &= g \\
 \Rightarrow f^{-1}(y) &= \frac{2 y}{1-y}, y \neq 1
 \end{aligned}$$

Q

EX1.3

-7

Invertible
linear
function $4x+3$



Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 4x+3$.

Show that f is invertible.

Find the inverse of f .

Given

$f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 4x+3$

For one-one

$$\begin{aligned}
 f(x) &= f(y) \\
 \Rightarrow 4 x+3 &= 4 y+3 \\
 \Rightarrow 4 x &= 4 y \\
 \Rightarrow x &= y
 \end{aligned}$$

$\therefore f$ is a one-one function.

Substitute

For onto

$$\begin{aligned}
 y \in \mathbb{R}, \text{ let } y &= 4 x+3 \\
 \Rightarrow x &= \frac{y-3}{4} \in \mathbb{R}
 \end{aligned}$$

Therefore, for any $y \in \mathbb{R}$, there exists $x = \frac{y-3}{4} \in \mathbb{R}$ such that

$$f(x) = f\left(\frac{y-3}{4}\right) = 4\left(\frac{y-3}{4}\right)+3 = y$$

$\therefore f$ is onto.

Conclusion

Thus, f is one-one and onto and therefore, f^{-1} exists.

Let us define $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = \frac{y-3}{4}$

Now,

$$(g \circ f)(x) = g(f(x)) = g(4x+3) = \frac{(4x+3)-3}{4} = x$$

$$(f \circ g)(y) = f(g(y)) = f\left(\frac{y-3}{4}\right) = 4\left(\frac{y-3}{4}\right) + 3 = y - 3 + 3 = y$$

$$\therefore g \circ f = f \circ g = I_{\mathbb{R}}$$

Hence, f is invertible and the inverse of f is given by

$$f^{-1}(y) = g(y) = \frac{y-3}{4}.$$

Q

EX1.3

-8

Invertible
function
 x^2+4 on
non-negative
reals



Consider $f: \mathbb{R}_+ \rightarrow [4, \infty)$ given by $f(x) = x^2 + 4$.

Show that f is invertible with inverse f^{-1} of given f by $f^{-1}(y) = \sqrt{y-4}$, where $\mathbb{R}_+$ is the set of all non-negative real numbers.

Given

$f: \mathbb{R}_+ \rightarrow [4, \infty)$ given by $f(x) = x^2 + 4$

For one-one:

$$\text{Let } f(x) = f(y)$$

$$\Rightarrow x^2 + 4 = y^2 + 4$$

$$\Rightarrow x^2 = y^2$$

$$\Rightarrow x = y \text{ [as } x \in \mathbb{R} \text{]}$$

$\therefore f$ is a one-one function.

Substitute

For onto:

For $y \in [4, \infty)$, let $y = x^2 + 4$

$$\Rightarrow x^2 = y - 4 \geq 0 \text{ [as } y \geq 4 \text{]}$$

$$\Rightarrow x = \sqrt{y-4} \geq 0$$

Therefore, for any $y \in \mathbb{R}$, there exists $x = \sqrt{y-4} \in \mathbb{R}$ such that

$$f(x) = f(\sqrt{y-4}) = (\sqrt{y-4})^2 + 4 = y - 4 + 4 = y$$

$\therefore f$ is an onto function.

Thus, f is one-one and onto and therefore, f^{-1} exists.

Conclusion

Let us define $g: [4, \infty) \rightarrow \mathbb{R}_+$ by

$$g(y) = \sqrt{y-4}$$

$$\text{Now, } g \circ f(x) = g(f(x)) = g(x^2 + 4) = \sqrt{(x^2 + 4) - 4} = \sqrt{x^2} = x$$

$$\text{And } f \circ g(y) = f(g(y)) = f(\sqrt{y-4}) = (\sqrt{y-4})^2 + 4 = (y-4) + 4 = y$$

$$\therefore g \circ f = f \circ g = I_{\mathbb{R}}$$

Hence, f is invertible and the inverse of f is given by

$$f^{-1}(y) = g(y) = \sqrt{y-4}.$$

Q

EX1.3

-9

Invertible
quadratic
function
 $9x^2+6x-5$ Consider $f: \mathbb{R}_+ \rightarrow [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$.Show that f is invertible with $f^{-1}(y) = \left(\frac{(\sqrt{y+6})-1}{3} \right)$.

Given

 $f: \mathbb{R}_+ \rightarrow [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$ Let y be an arbitrary element of $[-5, \infty)$.Let $y = 9x^2 + 6x - 5$

$$\Rightarrow y = (3x+1)^2 - 1 - 5$$

$$\Rightarrow y = (3x+1)^2 - 6$$

$$\Rightarrow (3x+1)^2 = y+6$$

$$\Rightarrow 3x+1 = \sqrt{y+6} \text{ [as } y \geq -5 \Rightarrow y+6 > 0 \text{]}$$

$$\Rightarrow x = \frac{\sqrt{y+6}-1}{3}$$

 $\therefore f$ is onto, thereby range $f = [-5, \infty)$.Let us define $g: [-5, \infty) \rightarrow \mathbb{R}_+$ as $g(y) = \frac{\sqrt{y+6}-1}{3}$

Conclusion

We have,

$$(g \circ f)(x) = g(f(x)) = g(9x^2 + 6x - 5)$$

$$= g((3x+1)^2 - 6)$$

$$= \frac{\sqrt{(3x+1)^2 - 6 + 6} - 1}{3}$$

$$= \frac{3x+1-1}{3} = x$$

And,

$$(f \circ g)(y) = f(g(y)) = f\left(\frac{\sqrt{y+6}-1}{3}\right)$$

$$= \left[3\left(\frac{\sqrt{y+6}-1}{3}\right) + 1\right]^2 - 6$$

$$= (\sqrt{y+6})^2 - 6 = y + 6 - 6 = y$$

Hence, f is invertible and the inverse of f is given by

$$f^{-1}(y) = g(y) = \frac{\sqrt{y+6}-1}{3}.$$

Q

EX1.3

-10

Uniqueness
of the
inverse
function

()

Let $f: X \rightarrow Y$ be an invertible function.Show that f has unique inverse.(Hint: suppose g_1 and g_2 are two inverses of f .)Then for all $y \in Y$, $f \circ g(y) = I_Y(y) = f \circ g(y)$.Use one-one ness of f .

Conclusion

Let $f: X \rightarrow Y$ be an invertible function.Also suppose f has two inverses (g_1 and g_2)Then, for all $y \in Y$,

$$f \circ g_1(y) = I_Y(y) = f \circ g_2(y)$$

$$\Rightarrow f(g_1(y)) = f(g_2(y))$$

$$\Rightarrow g_1(y) = g_2(y) \text{ [} f \text{ is invertible } \Rightarrow f \text{ is one-one]}$$

$$\Rightarrow g_1 = g_2 \text{ [} g \text{ is one-one]}$$

Hence, f has unique inverse.

Q

EX1.3

-11

Inverse of
the inverse -
finite
function

Consider $f: \{1, 2, 3\} \rightarrow \{a, b, c\}$ given by $f(1) = a, f(2) = b, f(3) = c$.

Find $(f^{-1})^{-1} = f$.

Given

Function $f: \{1, 2, 3\} \rightarrow \{a, b, c\}$ given by $f(1) = a, f(2) = b, f(3) = c$

If we define $g: \{a, b, c\} \rightarrow \{1, 2, 3\}$ as $g(a) = 1, g(b) = 2, g(c) = 3$

$$(f \circ g)(a) = f(g(a)) = f(1) = a$$

$$(f \circ g)(b) = f(g(b)) = f(2) = b$$

$$(f \circ g)(c) = f(g(c)) = f(3) = c$$

And,

$$(g \circ f)(1) = g(f(1)) = g(a) = 1$$

$$(g \circ f)(2) = g(f(2)) = g(b) = 2$$

$$(g \circ f)(3) = g(f(3)) = g(c) = 3$$

$$\therefore g \circ f = I_X \text{ and } f \circ g = I_Y \text{ where } X = \{1, 2, 3\} \text{ and } Y = \{a, b, c\}$$

Thus, the inverse of f exists and $f^{-1} = g$.

$$\therefore f^{-1}: \{a, b, c\} \rightarrow \{1, 2, 3\} \text{ is given by, } f^{-1}(a) = 1, f^{-1}(b) = 2, f^{-1}(c) = 3$$

Conclusion

We need to find the inverse of f^{-1} i.e., inverse of g .

If we define $h: \{1, 2, 3\} \rightarrow \{a, b, c\}$ as $h(1) = a, h(2) = b, h(3) = c$

$$(g \circ h)(1) = g(h(1)) = g(a) = 1$$

$$(g \circ h)(2) = g(h(2)) = g(b) = 2$$

$$(g \circ h)(3) = g(h(3)) = g(c) = 3$$

And,

$$(h \circ g)(a) = h(g(a)) = h(1) = a$$

$$(h \circ g)(b) = h(g(b)) = h(2) = b$$

$$(h \circ g)(c) = h(g(c)) = h(3) = c$$

$$\therefore g \circ h = I_X \text{ and } h \circ g = I_Y \text{ where } X = \{1, 2, 3\} \text{ and } Y = \{a, b, c\}$$

Thus, the inverse of g exists and $g^{-1} = h \Rightarrow (f^{-1})^{-1} = h$.

It can be noted that $h = f$.

$$\text{Hence, } (f^{-1})^{-1} = f$$

Q

EX1.3

-12

Inverse of
the inverse -
general proof

Let $f: X \rightarrow Y$ be an invertible function.

Show that the inverse of f^{-1} is f i.e., $(f^{-1})^{-1} = f$.

Conclusion

Let $f: X \rightarrow Y$ be an invertible function.

Then there exists a function $g: Y \rightarrow X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$

$$\text{Here, } f^{-1} = g$$

$$\text{Now, } g \circ f = I_X \text{ and } f \circ g = I_Y$$

$$\Rightarrow f^{-1} \circ f = I_X \text{ and } f \circ f^{-1} = I_Y$$

$$\text{Hence, } f^{-1}: Y \rightarrow X \text{ is invertible and } f^{-1} \text{ is f i.e., } (f^{-1})^{-1} = f.$$

Q

EX1.3

-13

MCQ -
composition
 $(3-x^3)^{1/3}$ 

If $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (3-x^3)^{1/3}$, then $f \circ f(x)$ is:

A.

$$\frac{1}{x^3}$$

B.

$$x^3$$

C.

$$x$$

D.

$$(3-x^3)$$

Conclusion

$f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (3-x^3) \frac{1}{3}$

$$f(x) = (3-x^3) \frac{1}{3}$$

$$\therefore f \circ f(x) = f(f(x)) = f\left((3-x^3) \frac{1}{3}\right) = \left[3 - \left((3-x^3) \frac{1}{3}\right)^3\right] \frac{1}{3}$$

$$= [3 - (3-x^3)] \frac{1}{3} = (x^3) \frac{1}{3} = x$$

$$\therefore f \circ f(x) = x$$

The correct answer is C.

C //

Q

EX1.3
-14

MCQ -
inverse of a
rational
function

()

If $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x+4}$.

The inverse of f is the map $g: \text{Range } f \rightarrow \mathbb{R} - \{-\frac{4}{3}\}$ given by :

A.

$$g(y) = \frac{3y}{3-4y}$$

B.

$$g(y) = \frac{4y}{4-3y}$$

C.

$$g(y) = \frac{4y}{3-4y}$$

D.

$$g(y) = \frac{3y}{4-3y}$$

Given

It is given that $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ is defined as $f(x) = \frac{4x}{3x+4}$

Let y be an arbitrary element of Range f .

Then, there exists $x \in \mathbb{R} - \{-\frac{4}{3}\}$ such that $y = f(x)$.

$$\Rightarrow y = \frac{4x}{3x+4}$$

$$\Rightarrow 3xy + 4y = 4x$$

$$\Rightarrow x(4-3y) = 4y$$

$$\Rightarrow x = \frac{4y}{4-3y}$$

Define $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ as $g(y) = \frac{4y}{4-3y}$

Conclusion

Now,

$$(g \circ f)(x) = g(f(x)) = g\left(\frac{4x}{3x+4}\right)$$

$$= \frac{4\left(\frac{4x}{3x+4}\right)}{4-3\left(\frac{4x}{3x+4}\right)} = \frac{16x}{12x+16-12x}$$

$$= \frac{16x}{16} = x$$

And

$$\begin{aligned}
 (f \circ g)(x) &= (g(x)) = f\left(\frac{4y}{4-3y}\right) \\
 &= \frac{4\left(\frac{4y}{4-3y}\right)}{3\left(\frac{4y}{4-3y}\right)+4} = \frac{16y}{12y+16-12y} \\
 &= \frac{16y}{16} = y \\
 \therefore g \circ f &= I_{\mathbb{R}-\{-\frac{4}{3}\}} \text{ and } f \circ g = I_{\text{Range } f}
 \end{aligned}$$

Thus, g is the inverse of f i.e., $f^{-1} = g$

Hence, the inverse of f is the map $g: \text{Range } f \rightarrow \mathbb{R}-\{-\frac{4}{3}\}$, which is given by $g(y) = \frac{4y}{4-3y}$.

The correct answer is B.

B

Q

EX1.4
-1

Checking
binary
operation
validity



Determine whether or not each of the definition of $*$ given below gives a binary operation.

In the event that $*$ is not a binary operation, give justification for this.

i.

On $\mathbb{Z}^+$, define $*$ by $a * b = a - b$

ii.

On $\mathbb{Z}^+$, define $*$ by $a * b = a b$

iii.

On $\mathbb{R}$, define $*$ by $a * b = a b^2$

iv.

On $\mathbb{Z}^+$, define $*$ by $a * b = |a - b|$

v.

On $\mathbb{Z}^+$, define $*$ by $a * b = a$

Given



i.

On $\mathbb{Z}^+$, define $*$ by $a * b = a - b$

It is not a binary operation as the image of $(1, 2)$ under $*$ is

$$1 * 2 = 1 - 2$$

$$\Rightarrow -1 \notin \mathbb{Z}^+.$$

Therefore, $*$ is not a binary operation.

Substitute

ii.

On $\mathbb{Z}^+$, define $*$ by $a * b = a b$

It is seen that for each $a, b \in \mathbb{Z}^+$, there is a unique element $a b$ in $\mathbb{Z}^+$.

This means that $*$ carries each pair (a, b) to a unique element $a * b = a b$ in $\mathbb{Z}^+$.

Therefore, $*$ is a binary operation.

iii.

On $\mathbb{R}$, define $*$ by $a * b = a b^2$

It is seen that for each $a, b \in \mathbb{R}$, there is a unique element $a b^2$ in $\mathbb{R}$.

This means that $*$ carries each pair (a, b) to a unique element $a * b = a b^2$ in $\mathbb{R}$.

Therefore, $*$ is a binary operation.

iv.

On $\mathbb{Z}^+$, define $*$ by $a * b = |a - b|$

It is seen that for each $a, b \in \mathbb{Z}^+$, there is a unique element $|a-b|$ in $\mathbb{Z}^+$.

This means that $*$ carries each pair (a, b) to a unique element $a * b = |a-b|$ in $\mathbb{Z}^+$.

Therefore, $*$ is a binary operation.

Conclusion

v.

On $\mathbb{Z}^+$, define $*$ by $a * b = a$

$*$ carries each pair (a, b) to a unique element in $a * b = a$ in $\mathbb{Z}^+$.

Therefore, $*$ is a binary operation.

Q

EX1.4

-2

Commutativity
and
associativity
of operations



For each binary operation $*$ defined below, determine whether $*$ is commutative or associative.

i.

On $\mathbb{Z}^+$, define $a * b = a-b$

ii.

On $\mathbb{Q}$, define $a * b = a b + 1$

iii.

On $\mathbb{Q}$, define $a * b = \frac{a b}{2}$

iv.

On $\mathbb{Z}^+$, define $a * b = 2^{a b}$

v.

On $\mathbb{Z}^+$, define $a * b = a^b$

vi.

On $\mathbb{R} - \{-1\}$, define $a * b = \frac{a}{b+1}$

Given

i.

On $\mathbb{Z}^+$, define $a * b = a-b$

It can be observed that $1 * 2 = 1-2 = -1$ and $2 * 1 = 2-1 = 1$.

$\therefore 1 * 2 \neq 2 * 1$; where $1, 2 \in \mathbb{Z}$

Hence, the operation $*$ is not commutative.

Also,

$$(1 * 2) * 3 = (1-2) * 3 = -1 * 3 = -1-3 = -4$$

$$1 * (2 * 3) = 1 * (2-3) = 1 * -1 = 1-(-1) = 2$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3)$$

where $1, 2, 3 \in \mathbb{Z}$

Hence, the operation $*$ is not associative.

Substitute

ii.

On $\mathbb{Q}$, define $a * b = a b + 1$

$$a b = b a \text{ for all } a, b \in \mathbb{Q}$$

$$\Rightarrow a b + 1 = b a + 1 \text{ for all } a, b \in \mathbb{Q}$$

$$\Rightarrow a * b = b * a \text{ for all } a, b \in \mathbb{Q}$$

Hence, the operation $*$ is commutative.

$$(1 * 2) * 3 = (1 \times 2 + 1) * 3 = 3 * 3 = 3 \times 3 + 1 = 10$$

$$1 * (2 * 3) = 1 * (2 \times 3 + 1) = 1 * 7 = 1 \times 7 + 1 = 8$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3)$$

where $1, 2, 3 \in \mathbb{Q}$

Hence, the operation $*$ is not associative.

iii.

On $\mathbb{Q}$, define $a * b = \frac{a b}{2}$

$$a \cdot b = b \cdot a \text{ for all } a, b \in \mathbb{Q}$$

$$\Rightarrow \frac{a \cdot b}{2} = \frac{a \cdot b}{2} \text{ for all } a, b \in \mathbb{Q}$$

$$\Rightarrow a * b = b * a \text{ for all } a, b \in \mathbb{Q}$$

Hence, the operation $*$ is commutative.

$$(a * b) * c = \left(\frac{a \cdot b}{2} \right) * c = \frac{\left(\frac{a \cdot b}{2} \right) \cdot c}{2} = \frac{a \cdot b \cdot c}{4}$$

And

$$a * (b * c) = a * \left(\frac{b \cdot c}{2} \right) = \frac{a \cdot \left(\frac{b \cdot c}{2} \right)}{2} = \frac{a \cdot b \cdot c}{4}$$

$$\therefore (a * b) * c = a * (b * c)$$

where $a, b, c \in \mathbb{Q}$

Hence, the operation $*$ is associative.

iv.

On $\mathbb{Z}^+$, define $a * b = 2^{a \cdot b}$

$$a \cdot b = b \cdot a \text{ for all } a, b \in \mathbb{Z}$$

$$\Rightarrow 2^{a \cdot b} = 2^{b \cdot a} \text{ for all } a, b \in \mathbb{Z}$$

$$\Rightarrow a * b = b * a \text{ for all } a, b \in \mathbb{Z}$$

Hence, the operation $*$ is commutative.

$$(1 * 2) * 3 = 2^{1 \times 2} * 3 = 4 * 3 = 2^{4 \times 3} = 2^{12}$$

$$1 * (2 * 3) = 1 * 2^{2 \times 3} = 1 * 2^6 = 1 * 64 = 2^{64}$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3)$$

where $1, 2, 3 \in \mathbb{Z}^+$

Hence, the operation $*$ is not associative.

v.

On $\mathbb{Z}^+$, define $a * b = a^b$

$$1 * 2 = 1^2 = 1$$

$$2 * 1 = 2^1 = 2$$

$$\therefore 1 * 2 \neq 2 * 1$$

where $1, 2 \in \mathbb{Z}^+$

Hence, the operation $*$ is not commutative.

$$(2 * 3) * 4 = 2^3 * 4 = 8 * 4 = 8^4 = 2^{12}$$

$$2 * (3 * 4) = 2 * 3^4 = 2 * 81 = 2^{81}$$

$$\therefore (2 * 3) * 4 \neq 2 * (3 * 4)$$

where $2, 3, 4 \in \mathbb{Z}^+$

Hence, the operation $*$ is not associative.

Conclusion

vi.

On $\mathbb{R} - \{-1\}$, define $a * b = \frac{a}{b+1}$

$$1 * 2 = \frac{1}{2+1} = \frac{1}{3}$$

$$2 * 1 = \frac{2}{1+1} = \frac{2}{2} = 1$$

$$\therefore 1 * 2 \neq 2 * 1$$

where $1, 2 \in \mathbb{R} - \{-1\}$

Hence, the operation $*$ is not commutative.

$$(1 * 2) * 3 = \frac{1}{3} * 3 = \frac{\frac{1}{3}}{3+1} = \frac{1}{12}$$

$$1 * (2 * 3) = 1 * \frac{2}{3+1} = 1 * \frac{2}{4} = 1 * \frac{1}{2} = \frac{1}{\frac{1}{2} + 1} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3)$$

where $1, 2, 3 \in \mathbb{R} - \{-1\}$

Hence, the operation $*$ is not associative.

Q

EX1.4

-3

Operation
table -
minimum
function



Consider the binary operation $\wedge$ on the set $\{1, 2, 3, 4, 5\}$ defined by $a \wedge b = \min\{a, b\}$.

Write the operation table of the operation $\wedge$.

Conclusion

The binary operation $\wedge$ on the set $\{1, 2, 3, 4, 5\}$ is defined by $a \wedge b = \min\{a, b\}$ for all $a, b \in \{1, 2, 3, 4, 5\}$.

The operation table for the given operation $\wedge$ can be given as:

Q

EX1.4

-4

Operation
table - given
multiplication
table

(())

Consider a binary operation $*$ on the set $\{1, 2, 3, 4, 5\}$ given by the following multiplication table.

i.

Compute $(2 * 3) * 4$ and $2 * (3 * 4)$

ii.

Is $*$ commutative?

iii.

Compute $(2 * 3) * (4 * 5)$.

(Hint: Use the following table)

Given

$$(2 * 3) * 4 = 1 * 4 = 1$$

i.

$$2 * (3 * 4) = 2 * 1 = 1$$

Substitute

ii.

For every $a, b \in \{1, 2, 3, 4, 5\}$, we have $a * b = b * a$.

Therefore, $*$ is commutative.

Conclusion

iii.

$$(2 * 3) * (4 * 5)$$

$$(2 * 3) = 1 \text{ and } (4 * 5) = 1$$

$$\therefore (2 * 3) * (4 * 5) = 1 * 1 = 1$$

Q

EX1.4

-5

Comparing
HCF
operation to
a given table



Let $*'$ be the binary operation on the set $\{1, 2, 3, 4, 5\}$ defined by $a *' b = \text{H.C.F. of } a \text{ and } b$.

Is the operation $*'$ same as the operation $*$ defined in Exercise 4 above?

Justify your answer.

Given

The binary operation on the set $\{1, 2, 3, 4, 5\}$ is defined by $a *' b = \text{H.C.F. of } a \text{ and } b$.

The operation table for the operation $*'$ can be given as:

Conclusion

The operation table for the operations $*$ ' and $*$ are same.

operation $*$ ' is same as operation $*$.

Q**EX1.4****-6**

LCM
operation -
properties
and identity



Let $*$ be the binary operation on N defined by $a * b = L.C.M.$ of a and b .

Find

i.

$$5 * 7, 20 * 16$$

ii.

Is $*$ commutative?

iii.

Is $*$ associative?

iv.

Find the identity of $*$ in N

v.

Which elements of N are invertible for the operation $*$?

Given

The binary operation on N is defined by $a * b = L.C.M.$ of a and b .

**Substitute**

i.

$$5 * 7 = L.C.M \text{ of } 5 \text{ and } 7 = 35$$

$$20 * 16 = LCM \text{ of } 20 \text{ and } 16 = 80$$

ii.

$L.C.M.$ of a and $b = LCM$ of b and a for all $a, b \in N$

$$\therefore a * b = b * a$$

Operation $*$ is commutative.

iii.

For $a, b, c \in N$

$$(a * b) * c = (L.C.M. \text{ of } a \text{ and } b) * c = L.C.M. \text{ of } a, b, c$$

$$a * (b * c) = a * (L.C.M. \text{ of } b \text{ and } c) = L.C.M. \text{ of } a, b, c$$

$$\therefore (a * b) * c = a * (b * c)$$

Operation $*$ is associative.

iv.

$L.C.M.$ of a and $1 = a = L.C.M.$ of 1 and a for all $a \in N$

$$a * 1 = a = 1 * a \text{ for all } a \in N$$

Therefore, 1 is the identity of $*$ in N .

Conclusion

v.

An element a in N is invertible with respect to the operation $*$ if there exists an element b in N , such that $a * b = e = b * a$

$$e = 1$$

$L.C.M.$ of a and $b = 1 = LCM$ of b and a possible only when a and b are equal to 1 .

1 is the only invertible element of N with respect to the operation $*$.

Q

EX1.4

-7

LCM on a finite set is not a binary operation



Is $*$ defined on the set $\{1,2,3,4,5\}$ by $a * b = \text{LCM of } a \text{ and } b$ a binary operation? Justify your answer.

Given

The operation $*$ on the set $\{1,2,3,4,5\}$ is defined by $a * b = \text{LCM of } a \text{ and } b$.

The operation table for the operation $*$ can be given as:

Conclusion

$3 * 2 = 2 * 3 = 6 \notin A$,
 $5 * 2 = 2 * 5 = 10 \notin A$,
 $3 * 4 = 4 * 3 = 12 \notin A$,
 $3 * 5 = 5 * 3 = 15 \notin A$,
 $4 * 5 = 5 * 4 = 20 \notin A$

The given operation $*$ is not a binary operation.

Q

EX1.4

-8

HCF operation - commutative, associative, no identity

()

Let $*$ be the binary operation on N defined by $a * b = \text{H.C.F. of } a \text{ and } b$.

Is $*$ commutative?

Is $*$ associative?

Does there exist identity for this binary operation on N ?

Given

The binary operation $*$ on N defined by $a * b = \text{H.C.F. of } a \text{ and } b$.

Substitute

$\therefore a * b = b * a$

Operation $*$ is commutative.

For all $a, b, c \in N$,

$(a * b) * c = (\text{HCF of } a \text{ and } b) * c = \text{HCF of } a, b, c$

$a * (b * c) = a * (\text{HCF of } b \text{ and } c) = \text{HCF of } a, b, c$

$\therefore (a * b) * c = a * (b * c)$

Operation $*$ is associative.

Conclusion

$e \in N$ will be the identity for the operation $*$ if $a * e = a = e * a$ for all $a \in N$.

But this relation is not true for any $a \in N$.

Operation $*$ does not have any identity in N .

Q

EX1.4

-9

Commutativity and associativity on Q



Let $*$ be the binary operation on Q of rational numbers as follows:

i.

$a * b = a - b$

ii.

$a * b = a^2 + b^2$

iii.

$a * b = a + a b$

iv.

$a * b = (a - b)^2$

v.

$a * b = \frac{a b}{4}$

vi.

$a * b = a b^2$

Find which of the binary operations are commutative and which are associative.

Given

i.



On $\mathbb{Q}$, the operation $*$ is defined as $a * b = a - b$

$$\frac{1}{2} * \frac{1}{3} = \frac{1}{2} - \frac{1}{3} = \frac{3-2}{3} = \frac{1}{3}$$

And

$$\frac{1}{3} * \frac{1}{2} = \frac{1}{3} - \frac{1}{2} = \frac{2-3}{6} = \frac{-1}{6}$$

$$\therefore \left(\frac{1}{2} * \frac{1}{3} \right) \neq \left(\frac{1}{3} * \frac{1}{2} \right)$$

where $\frac{1}{2}, \frac{1}{3} \in \mathbb{Q}$

Operation $*$ is not commutative.

$$\left(\frac{1}{2} * \frac{1}{3} \right) * \frac{1}{4} = \left(\frac{1}{2} - \frac{1}{3} \right) * \frac{1}{4} = \frac{1}{6} * \frac{1}{4} = \frac{1}{6} - \frac{1}{4} = \frac{2-3}{12} = \frac{-1}{12}$$

$$\frac{1}{2} * \left(\frac{1}{3} * \frac{1}{4} \right) = \frac{1}{2} * \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{2} * \frac{1}{12} = \frac{1}{2} - \frac{1}{12} = \frac{6-1}{12} = \frac{5}{12}$$

$$\therefore \left(\frac{1}{2} * \frac{1}{3} \right) * \frac{1}{4} \neq \frac{1}{2} * \left(\frac{1}{3} * \frac{1}{4} \right)$$

Operation $*$ is not associative.

Substitute

ii.

On $\mathbb{Q}$, the operation $*$ is defined as $a * b = a^2 + b^2$

For $a, b \in \mathbb{Q}$

$$a * b = a^2 + b^2 = b^2 + a^2 = b * a$$

$$\therefore a * b = b * a$$

Operation $*$ is commutative.

$$(1 * 2) * 3 = (1^2 + 2^2) * 3 = (1+4) * 3 = 5 * 3 = 5^2 + 3^2 = 25+9 = 34$$

$$1 * (2 * 3) = 1 * (2^2 + 3^2) = 1 * (4+9) = 1 * 13 = 1^2 + 13^2 = 1+169 = 170$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3) \text{ where } 1, 2, 3 \in \mathbb{Q}$$

Operation $*$ is not associative.

iii.

On $\mathbb{Q}$, the operation $*$ is defined as $a * b = a + a b$

$$1 * 2 = 1 + 1 \times 2 = 1 + 2 = 3$$

$$2 * 1 = 2 + 2 \times 1 = 2 + 2 = 4$$

$$\therefore 1 * 2 \neq 2 * 1$$

where $1, 2 \in \mathbb{Q}$

Operation $*$ is not commutative.

$$(1 * 2) * 3 = (1 + 1 \times 2) * 3 = 3 * 3 = 3 + 3 \times 3 = 3 + 9 = 12$$

$$1 * (2 * 3) = 1 * (2 + 2 \times 3) = 1 * 8 = 1 + 1 \times 8 = 1 + 8 = 9$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3) \text{ where } 1, 2, 3 \in \mathbb{Q}$$

Operation $*$ is not associative.

iv.

On $\mathbb{Q}$, the operation $*$ is defined as $a * b = (a-b)^2$

For $a, b \in \mathbb{Q}$

$$a * b = (a-b)^2$$

$$b * a = (b-a)^2 = [-(a-b)]^2 = (a-b)^2$$

$$\therefore a * b = b * a$$

Operation $*$ is commutative.

$$(1 * 2) * 3 = (1-2)^2 * 3 = (-1)^2 * 3 = 1 * 3 = (1-3)^2 = (-2)^2 = 4$$

$$1 * (2 * 3) = 1 * (2-3)^2 = 1 * (-1)^2 = 1 * 1 = (1-1)^2 = 0$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3) \text{ where } 1, 2, 3 \in \mathbb{Q}$$

Operation $*$ is not associative.

v.

On $\mathbb{Q}$, the operation $*$ is defined as $a * b = \frac{a b}{4}$

For $a, b \in \mathbb{Q}$

$$a * b = \frac{a b}{4} = \frac{b a}{4} = b * a$$

$$\therefore a * b = b * a$$

Operation $*$ is commutative.

For $a, b, c \in \mathbb{Q}$

$$(a * b) * c = \frac{\frac{a b}{4}}{4} * c = \frac{\frac{a b}{4} \cdot c}{4} = \frac{a b c}{16}$$

$$a * (b * c) = a * \frac{b c}{4} = \frac{a \cdot \frac{b c}{4}}{4} = \frac{a b c}{16}$$

$$\therefore (a * b) * c = a * (b * c)$$

where $a, b, c \in \mathbb{Q}$

Operation $*$ is associative.

vi.

On $\mathbb{Q}$, the operation $*$ is defined as $a * b = a b^2$

$$\frac{1}{2} * \frac{1}{3} = \frac{1}{2} \cdot \left(\frac{1}{3}\right)^2 = \frac{1}{2} \cdot \frac{1}{9} = \frac{1}{18}$$

$$\frac{1}{3} * \frac{1}{2} = \frac{1}{3} \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

$$\therefore \left(\frac{1}{2} * \frac{1}{3}\right) \neq \left(\frac{1}{3} * \frac{1}{2}\right)$$

where $\frac{1}{2}, \frac{1}{3} \in \mathbb{Q}$

Operation $*$ is not commutative.

$$\left(\frac{1}{2} * \frac{1}{3}\right) * \frac{1}{4} = \left(\frac{1}{2} \cdot \left(\frac{1}{3}\right)^2\right) * \frac{1}{4} = \frac{1}{18} * \frac{1}{4} = \frac{1}{18} \cdot \left(\frac{1}{4}\right)^2 = \frac{1}{18 \times 16}$$

$$\frac{1}{2} * \left(\frac{1}{3} * \frac{1}{4}\right) = \frac{1}{2} * \left(\frac{1}{3} \cdot \left(\frac{1}{4}\right)^2\right) = \frac{1}{2} * \frac{1}{48} = \frac{1}{2} \cdot \left(\frac{1}{48}\right)^2 = \frac{1}{2 \times (48)^2}$$

$$\therefore \left(\frac{1}{2} * \frac{1}{3}\right) * \frac{1}{4} \neq \frac{1}{2} * \left(\frac{1}{3} * \frac{1}{4}\right)$$

Operation $*$ is not associative.

Conclusion

Operations defined in (ii), (iv), (v) are commutative and the operation defined in (v) is associative.

Q

EX1.4 -10

Identity
element
among
operations
on $\mathbb{Q}$



Find which of the operations given above has identity.

Conclusion

An element $e \in \mathbb{Q}$ will be the identity element for the operation $*$ if

$$a * e = a = e * a, \text{ for all } a \in \mathbb{Q}$$

$$a * b = \frac{a b}{4}$$

$$\Rightarrow a * e = a$$

$$\Rightarrow \frac{a e}{4} = a$$

$$\Rightarrow e = 4$$

Similarly, it can be checked for $e * a = a$, we get $e = 4$ is the identity.

Q

EX1.4
-11Componentwise
addition on N
 $\times N$ 

$A = N \times N$ and $*$ be the binary operation on A defined by $(a, b) * (c, d) = (a+c, b+d)$.

Show that $*$ is commutative and associative.

Find the identity element for $*$ on A , if any.

Given

$A = N \times N$ and $*$ be the binary operation on A defined by

Substitute

$$\begin{aligned}(a, b) * (c, d) &= (a+c, b+d) \\ (a, b) * (c, d) &\in A \\ a, b, c, d &\in N \\ (a, b) * (c, d) &= (a+c, b+d) \\ (c, d) * (a, b) &= (c+a, d+b) = (a+c, b+d) \\ \therefore (a, b) * (c, d) &= (c, d) * (a, b)\end{aligned}$$

Operation $*$ is commutative.

Now, let $(a, b), (c, d), (e, f) \in A$

$$\begin{aligned}a, b, c, d, e, f &\in N \\ [(a, b) * (c, d)] * (e, f) &= (a+c, b+d) * (e, f) = (a+c+e, b+d+f) \\ (a, b) * [(c, d) * (e, f)] &= (a, b) * (c+e, d+f) = (a+c+e, b+d+f) \\ \therefore [(a, b) * (c, d)] * (e, f) &= (a, b) * [(c, d) * (e, f)]\end{aligned}$$

Operation $*$ is associative.

Conclusion

An element $e = (e_1, e_2) \in A$ will be an identity element for the operation $*$ if

$$a + e = a = e * a \text{ for all } a = (a_1, a_2) \in A \text{ i.e., } (a_1 + e_1, a_2 + e_2) = (a_1, a_2) = (e_1 + a_1, e_2 + a_2),$$

which is not true for any element in A .

Therefore, the operation $*$ does not have any identity element.

Q

EX1.4
-12True or false
- binary
operation
statements

()

State whether the following statements are true or false.

Justify.

i.

For an arbitrary binary operation $*$ on a set N , $a * a = a$ for all $a \in N$.

ii.

If $*$ is a commutative binary operation on N , then $a * (b * c) = (c * b) * a$

Given

i.

Define operation $*$ on a set N as $a * a = a$ for all $a \in N$.

In particular, for $a = 3$,

$$3 * 3 = 9 \neq 3$$

Therefore, statement (i) is false.

Conclusion

ii.

$$\text{R.H.S.} = (c * b) * a$$

$$= (b * c) * a \text{ [} * \text{ is commutative]}$$

$$= a * (b * c) \text{ [Again, as } * \text{ is commutative]}$$

$$= \text{L.H.S.}$$

$$\therefore a * (b * c) = (c * b) * a$$

Therefore, statement (ii) is true.

Q

EX1.4
-13MCQ -
commutative
but not
associative
operation

Consider a binary operation $*$ on N defined as $a * b = a^3 + b^3$.

Choose the correct answer.

A.

Is $*$ both associative and commutative?

B.

Is $*$ commutative but not associative?

C.

Is $*$ associative but not commutative?

D.

Is $*$ neither commutative nor associative?

Given

On N , operation $*$ is defined as $a * b = a^3 + b^3$.



Substitute

For all $a, b \in N$

$$a * b = a^3 + b^3 = b^3 + a^3 = b * a$$

Operation $*$ is commutative.

$$(1 * 2) * 3 = (1^3 + 2^3) * 3 = (1 + 8) * 3 = 9 * 3 = 9^3 + 3^3 = 729 + 27 = 756$$

$$1 * (2 * 3) = 1 * (2^3 + 3^3) = 1 * (8 + 27) = 1 * 35 = 1^3 + 35^3 = 1 + 42875 = 42876$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3)$$

Operation $*$ is not associative.

Conclusion

Therefore, Operation $*$ is commutative, but not associative.

The correct answer is B.

B //

Q

MISC
-1Inverse of
 $f(x) = 10x + 7$ 

Let $f: R \rightarrow R$ be defined as $f(x) = 10x + 7$.

Find the function $g: R \rightarrow R$ such that $g \circ f = f \circ g = I_R$.

Given

$f: R \rightarrow R$ is defined as $f(x) = 10x + 7$



Substitute

For one-one:

$$f(x) = f(y) \text{ where } x, y \in R$$

$$\Rightarrow 10x + 7 = 10y + 7$$

$$\Rightarrow x = y$$

$\therefore f$ is one-one.

For onto:

$$y \in R, \text{ Let } y = 10x + 7$$

$$\Rightarrow x = \frac{y-7}{10} \in R$$

For any $y \in R$, there exists $x = \frac{y-7}{10} \in R$ such that

$$f(x) = f\left(\frac{y-7}{10}\right) = 10\left(\frac{y-7}{10}\right) + 7 = y - 7 + 7 = y$$

$\therefore f$ is onto.

Thus, f is an invertible function.

Conclusion

Let us define $g: R \rightarrow R$ as $g(y) = \frac{y-7}{10}$.

Now,

$$g \circ f(x) = g(f(x)) = g(10x+7) = \frac{(10x+7)-7}{10} = \frac{10x}{10} = 10$$

And,

$$f \circ g(y) = f(g(y)) = f\left(\frac{y-7}{10}\right) = 10\left(\frac{y-7}{10}\right) + 7 = y-7+7 = y$$

$$\therefore g \circ f = I_R \text{ and } f \circ g = I_R$$

$$\text{Hence, the required function } g: R \rightarrow R \text{ as } g(y) = \frac{y-7}{10}.$$

Q

MISC

-2

Invertible
function on
whole
numbers



Let $f: W \rightarrow W$ be defined as $f(n) = n-1$, if n is odd and $f(n) = n+1$, if n is even.

Show that f is invertible.

Find the inverse of f .

Here, W is the set of all whole numbers.

Given

$$f: W \rightarrow W \text{ is defined as } f(n) = \begin{cases} n-1, & \text{If } n \text{ is odd} \\ n+1, & \text{If } n \text{ is even} \end{cases}$$

Substitute

For one-one:

$$f(n) = f(m)$$

If n is odd and m is even, then we will have $n-1 = m+1$.

$$\Rightarrow n-m = 2$$

Similarly, the possibility of n being even and m being odd can also be ignored under a similar argument.

$\therefore$ Both n and m must be either odd or even.

Now, if both n and m are odd, then we have:

$$f(n) = f(m)$$

$$\Rightarrow n-1 = m-1$$

$$\Rightarrow n = m$$

Again, if both n and m are even, then we have:

$$f(n) = f(m)$$

$$\Rightarrow n+1 = m+1$$

$$\Rightarrow n = m$$

$\therefore f$ is one-one.

For onto:

Any odd number $2r+1$ in co-domain N is the image of $2r$ in domain N and any even number $2r$ in co-domain N is the image of $2r+1$ in domain N .

$\therefore f$ is onto.

f is an invertible function.

Conclusion

$$\text{Let us define } g: W \rightarrow W \text{ as } f(m) = \begin{cases} m-1, & \text{If } m \text{ is odd} \\ m+1, & \text{If } m \text{ is even} \end{cases}$$

When r is odd

$$g \circ f(n) = g(f(n)) = g(n-1) = n-1+1 = n$$

When r is even

$$g \circ f(n) = g(f(n)) = g(n+1) = n+1-1 = n$$

When m is odd

$$f \circ g(n) = f(g(m)) = f(m-1) = m-1+1 = m$$

When m is even

$$f \circ g(m) = f(g(m)) = f(m+1) = m+1-1 = m$$

$$\therefore g \circ f = I_W \text{ and } f \circ g = I_W$$

f is invertible and the inverse of f is given by $f^{-1} = g$, which is the same as f .

inverse of f is f itself.

Q

MISC
-3

Composition
 $f(f(x))$ for a
quadratic

(())

If $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^2 - 3x + 2$, find $f(f(x))$.

Conclusion

$f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = x^2 - 3x + 2$.

$$\begin{aligned} f(f(x)) &= f(x^2 - 3x + 2) \\ &= (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2 \\ &= (x^4 + 9x^2 + 4 - 6x^3 - 12x + 4x^2) + (-3x^2 + 9x - 6) + 2 \\ &= x^4 - 6x^3 + 10x^2 - 3x \end{aligned}$$

Q

MISC
-6

Injective
composition
with a non-
injective g

✧

Give examples of two functions $f: \mathbb{N} \rightarrow \mathbb{Z}$ and $g: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $g \circ f$ is injective but g is not injective.

(Hint: Consider $f(x) = x$ and $g(x) = |x|$)

Given

Define $f: \mathbb{N} \rightarrow \mathbb{Z}$ as $f(x) = x$ and $g: \mathbb{Z} \rightarrow \mathbb{Z}$ as $g(x) = |x|$

Substitute

Let us first show that g is not injective.

$$\begin{aligned} (-1) &= |-1| = 1 \\ (1) &= |1| = 1 \\ \therefore (-1) &= g(1), \text{ but } -1 \neq 1 \\ \therefore g &\text{ is not injective.} \end{aligned}$$

Conclusion

$g \circ f: \mathbb{N} \rightarrow \mathbb{Z}$ is defined as $g \circ f(x) = g(f(x)) = g(x) = |x|$

$x, y \in \mathbb{N}$ such that $g \circ f(x) = g \circ f(y)$

$$\Rightarrow |x| = |y|$$

Since $x, y \in \mathbb{N}$, both are positive.

$$\therefore |x| = |y|$$

$$\Rightarrow x = y$$

$\therefore g \circ f$ is injective.

Q

MISC
-7

Onto
composition
with a non-
onto f

✧

Given examples of two functions $f: \mathbb{N} \rightarrow \mathbb{N}$ and $g: \mathbb{N} \rightarrow \mathbb{N}$ such that $g \circ f$ is onto but f is not onto.

(Hint: Consider $f(x) = x + 1$ and $g(x) = \begin{cases} x-1, & \text{if } x > 1 \\ 1, & \text{if } x = 1 \end{cases}$)

Given

Define $f: \mathbb{N} \rightarrow \mathbb{N}$ as $f(x) = x + 1$ and $g: \mathbb{N} \rightarrow \mathbb{N}$ as $g(x) = \begin{cases} x-1, & \text{if } x > 1 \\ 1, & \text{if } x = 1 \end{cases}$

Substitute

Let us first show that g is not onto.

Consider element 1 in co-domain $\mathbb{N}$.

This element is not an image of any of the elements in domain $\mathbb{N}$.

$\therefore g$ is not onto.

Conclusion

$g: \mathbb{N} \rightarrow \mathbb{N}$ is defined by

$$g \circ f(x) = g(f(x)) = g(x+1) = x+1-1 = x \quad [x \in \mathbb{N} \Rightarrow x+1 > 1]$$

For $y \in \mathbb{N}$, there exists $x = y \in \mathbb{N}$ such that $g \circ f(x) = y$.

$\therefore g \circ f$ is onto.

Q

MISC
-9Identity and
invertible
element in a
power set

Given a non-empty set X , consider the binary operation $*$: $P(X) \times P(X) \rightarrow P(X)$ given by $A * B = A \cap B \forall A, B$ in $P(X)$ is the power set of X .
Show that X is the identity element for this operation and X is the only invertible element in $P(X)$ with respect to the operation $*$.

Given

 $P(X) \times P(X) \rightarrow P(X)$ given by $A * B = A \cap B \forall A, B$ in $P(X)$
 $A \cap X = A = X \cap A$ for all $A \in P(X)$
 $\Rightarrow A * X = A = X * A$ for all $A \in P(X)$
 X is the identity element for the given binary operation $*$.

Conclusion

 An element $A \in P(X)$ is invertible if there exists $B \in P(X)$ such that

 $A * B = X = B * A$ [As X is the identity element]

Or

 $A \cap B = X = B \cap A$

 This case is possible only when $A = X = B$.

 X is the only invertible element in $P(X)$ with respect to the given operation $*$.

Q

MISC
-11Inverse of a
function
between
finite sets

()

 Let $S = \{a, b, c\}$ and $T = \{1, 2, 3\}$.

 Find F^{-1} of the following functions F from S to T , if it exists.

i.

 $F = \{(a, 3), (b, 2), (c, 1)\}$

ii.

 $F = \{(a, 2), (b, 1), (c, 1)\}$

Given

 $S = \{a, b, c\}, T = \{1, 2, 3\}$

Substitute

i.

 $F: S \rightarrow T$ is defined by $F = \{(a, 3), (b, 2), (c, 1)\}$
 $\Rightarrow F(a) = 3, F(b) = 2, F(c) = 1$

 Therefore, $F^{-1}: T \rightarrow S$ is given by $F^{-1} = \{(3, a), (2, b), (1, c)\}$

Conclusion

ii.

 $F: S \rightarrow T$ is defined by $F = \{(a, 2), (b, 1), (c, 1)\}$

 Since, $F(b) = F(c) = 1$, F is not one-one.

 Hence, F is not invertible i.e., F^{-1} does not exist.

Q

MISC
-12Commutative,
associative,
distributive
operations
 Consider the binary operations $*$: $R \times R \rightarrow R$ and $\circ$: $R \times R \rightarrow R$ defined as $a * b = |a - b|$ and $a \circ b = a, \forall a, b \in R$.

 Show that $*$ is commutative but not associative $\circ$ is associative but not commutative.

 Further, show that $\forall a, b, c \in R, a * (b \circ c) = (a * b) \circ (a * c)$.

 [If it is so, we say that the operation $*$ distributes over the operation $\circ$].

 Does $\circ$ distribute over $*$?

Justify your answer.

Given

 It is given that $*$: $R \times R \rightarrow R$ and $\circ$: $R \times R \rightarrow R$ defined as $a * b = |a - b|$ and $a \circ b = a, \forall a, b \in R$.

 For $a, b \in R$, we have $a * b = |a - b|$ and $b * a = |b - a| = |-(a - b)| = |a - b|$

Substitute

$$\therefore a * b = b * a$$

$\therefore$ The operation $*$ is commutative.

$$(1 * 2) * 3 = (|1-2|) * 3 = 1 * 3 = |1-3| = 2$$

$$1 * (2 * 3) = 1 * (|2-3|) = 1 * 1 = |1-1| = 0$$

$$\therefore (1 * 2) * 3 \neq 1 * (2 * 3) \text{ where } 1, 2, 3 \in \mathbb{R}$$

$\therefore$ The operation $*$ is not associative.

Now, consider the operation $\circ$:

It can be observed that $1 \circ 2 = 1$ and $2 \circ 1 = 2$.

$$\therefore 1 \circ 2 \neq 2 \circ 1 \text{ (where } 1, 2 \in \mathbb{R} \text{)}$$

$\therefore$ The operation $\circ$ is not commutative.

Let $a, b, c \in \mathbb{R}$.

Then, we have:

$$(a \circ b) \circ c = a \circ c = a$$

$$a \circ (b \circ c) = a \circ b = a$$

$$\Rightarrow (a \circ b) \circ c = a \circ (b \circ c)$$

$\therefore$ The operation $\circ$ is associative.

Now, let $a, b, c \in \mathbb{R}$, then we have:

$$a * (b \circ c) = a * b = |a-b|$$

$$(a * b) \circ (a * c) = (|a-b|) \circ (|a-c|) = |a-b|$$

$$\text{Hence, } a * (b \circ c) = (a * b) \circ (a * c)$$

Conclusion

Now,

$$1 \circ (2 * 3) = 1 \circ (|2-3|) = 1 \circ 1 = 1$$

$$(1 \circ 2) * (1 \circ 3) = 1 * 1 = |1-1| = 0$$

$$\therefore 1 \circ (2 * 3) \neq (1 \circ 2) * (1 \circ 3) \text{ where } 1, 2, 3 \in \mathbb{R}$$

$\therefore$ The operation $\circ$ does not distribute over $*$.

Q

MISC
-13

Identity and
inverses
under
symmetric
difference



Given a non - empty set X , let $*$: $P(X) \times P(X) \rightarrow P(X)$ be defined as $A * B = (A-B) \cup (B-A)$, $\forall A, B \in P(X)$.

Show that the empty set Φ is the identity for the operation $*$ and all the elements A of $P(X)$ are invertible with $A^{-1} = A$.

(Hint: $(A-\Phi) \cup (\Phi-A) = A$ and $(A-A) \cup (A-A) = A * A = \Phi$).

Given

It is given that $*$: $P(X) \times P(X) \rightarrow P(X)$ is defined as $A * B = (A-B) \cup (B-A)$, $\forall A, B \in P(X)$ $A \in P(X)$ then,

Substitute

$$A * \Phi = (A-\Phi) \cup (\Phi-A) = A \cup \Phi = A$$

$$\Phi * A = (\Phi-A) \cup (A-\Phi) = \Phi \cup A = A$$

$$\therefore A * \Phi = A = \Phi * A \text{ for all } A \in P(X)$$

Φ is the identity for the operation $*$.

Conclusion

Element $A \in P(X)$ will be invertible if there exists $B \in P(X)$ such that $A * B = \Phi = B * A$ [As Φ is the identity element]

$$A * A = (A-A) \cup (A-A) = \Phi \cup \Phi = \Phi \text{ for all } A \in P(X) .$$

All the elements A of $P(X)$ are invertible with $A^{-1} = A$.

Q

MISC
-14Identity and
inverses -
modular
addition

Define a binary operation $*$ on the set $\{0,1,2,3,4,5\}$ as

$a+b$, if $a+b < 6$

$a+b-6$ if $a+b \geq 6$

Show that zero is the identity for this operation and each element $a \neq 0$ of the set is invertible with $6-a$ being the inverse of a .

Given

Let $X = \{0,1,2,3,4,5\}$

The operation $*$ is defined as $a+b = \begin{cases} a+b, & \text{if } a+b < 6 \\ a+b-6, & \text{if } a+b \geq 6 \end{cases}$

An element $e \in X$ is the identity element for the operation $*$, if $a * e = a = e * a \forall a \in X$

For $a \in X$,

Substitute

$$a * 0 = a+0 = a \quad [a \in X \Rightarrow a+0 < 6]$$

$$0 * a = 0+a = a \quad [a \in X \Rightarrow 0+a < 6]$$

$$\therefore a * 0 = a = 0 * a \quad \forall a \in X$$

Thus, 0 is the identity element for the given operation $*$.

Conclusion

An element $a \in X$ is invertible if there exists $b \in X$ such that $a * b = 0 = b * a$.

$$\text{i.e., } \begin{cases} a+b = 0 = b+a, & \text{if } a+b < 6 \\ a+b-6 = 0 = b+a-6 & \text{if } a+b \geq 6 \end{cases}$$

$$\Rightarrow a = -b \text{ or } b = 6-a$$

$$X = \{0,1,2,3,4,5\} \text{ and } a, b \in X.$$

Then $a \neq -b$.

$\therefore b = 6-a$ is the inverse of a for all $a \in X$.

Inverse of an element $a \in X$, $a \neq 0$ is $6-a$ i.e., $a^{-1} = 6-a$.

Q

MISC
-18Signum and
greatest
integer
composition

()

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the Signum Function defined as

1, $x > 0$

0, $x = 0$

-1, $x < 0$

greatest integer function given by $g(x) = [x]$, where $[x]$ is greatest integer less than or equal to x .

Then does $f \circ g$ and $g \circ f$ coincide in $(0,1)$?

Conclusion

It is given that $f: \mathbb{R} \rightarrow \mathbb{R}$ be the Signum Function defined as

1, $x > 0$

0, $x = 0$

-1, $x < 0$

Also $g: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $g(x) = [x]$, where $[x]$ is greatest integer less than or equal to x .

Now let $x \in (0,1]$,

$$[x] = 1 \text{ if } x = 1 \text{ and } [x] = 0 \text{ if } 0 < x < 1.$$

$$\therefore f \circ g(x) = f(g(x)) = f([x]) = \begin{cases} f(1), & \text{if } x = 1 \\ f(0), & \text{if } x \in (0,1) \end{cases} = \begin{cases} 1, & \text{if } x = 1 \\ 0, & \text{if } x \in (0,1) \end{cases}$$

$$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= g(1) \\ &= [1] = 1 \end{aligned}$$

Thus, when $x \in (0,1)$, we have $f \circ g(x) = 0$ and $g \circ f(x) = 1$.

Hence, $f \circ g$ and $g \circ f$ does not coincide in $(0,1)$.

Q

MISC
-19

MCQ -
number of
binary
operations
on {a,b}



Number of binary operations on the set {a, b} are

- A.
10
- B.
16
- C.
20
- D.
8

Conclusion

A binary operation $*$ on $\{a, b\}$ is a function from $\{a, b\} \times \{a, b\} \rightarrow \{a, b\}$ i.e., $*$ is a function from $\{(a, a), (a, b), (b, a), (b, b)\} \rightarrow \{a, b\}$

Hence, the total number of binary operations on the set $\{a, b\}$ is $2^4 = 16$.

The correct answer is B.

B

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